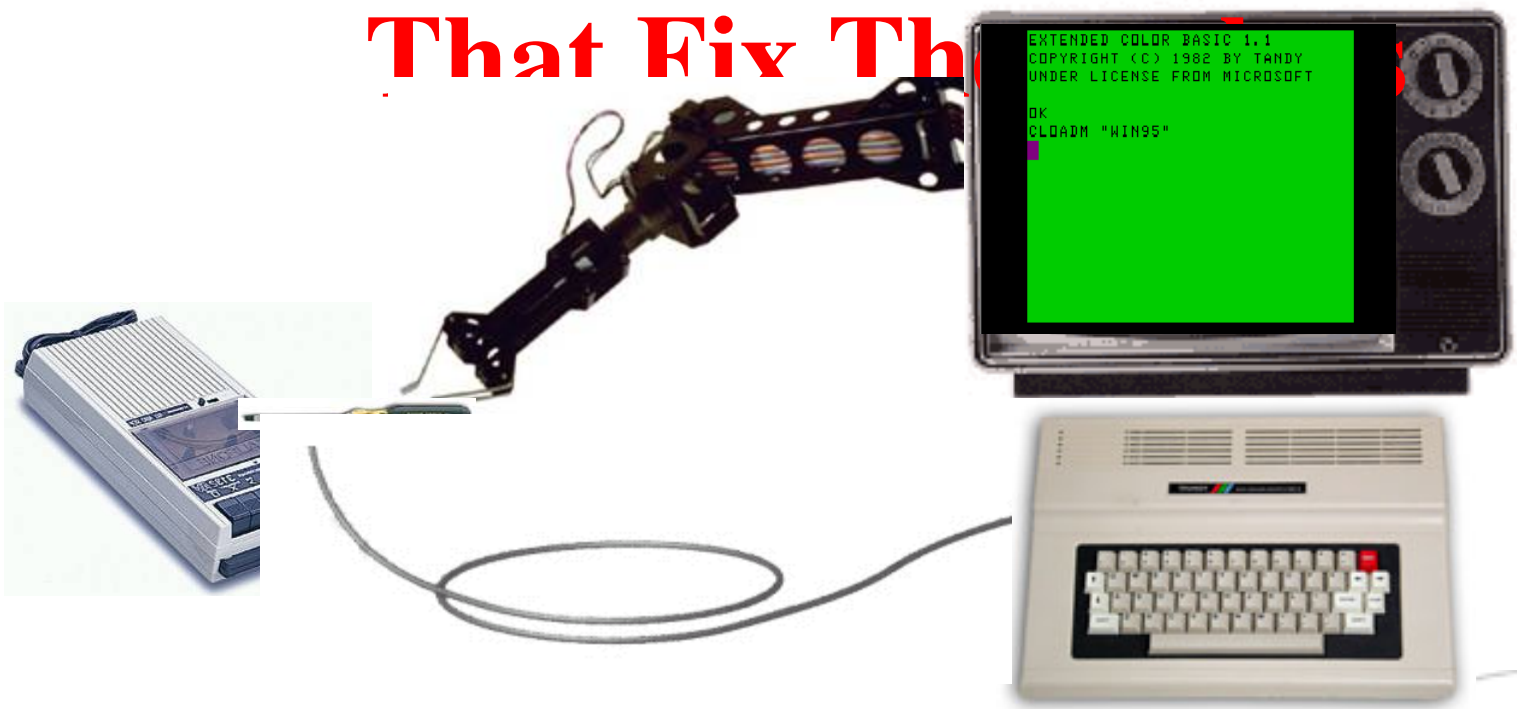
# sihT esraP oT yrT

# Quantum Computers That Fix Th

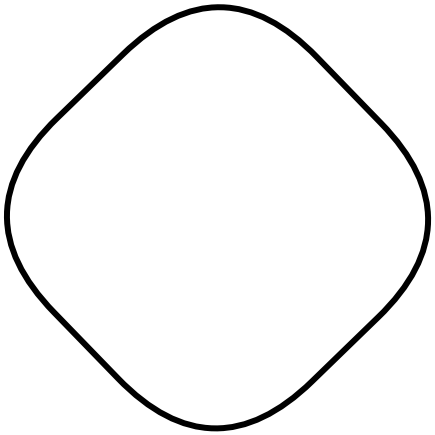



**Dave Bacon**
**Caltech**
Department of Physics
Institute for Quantum Information

# TRS-80 Color Computer

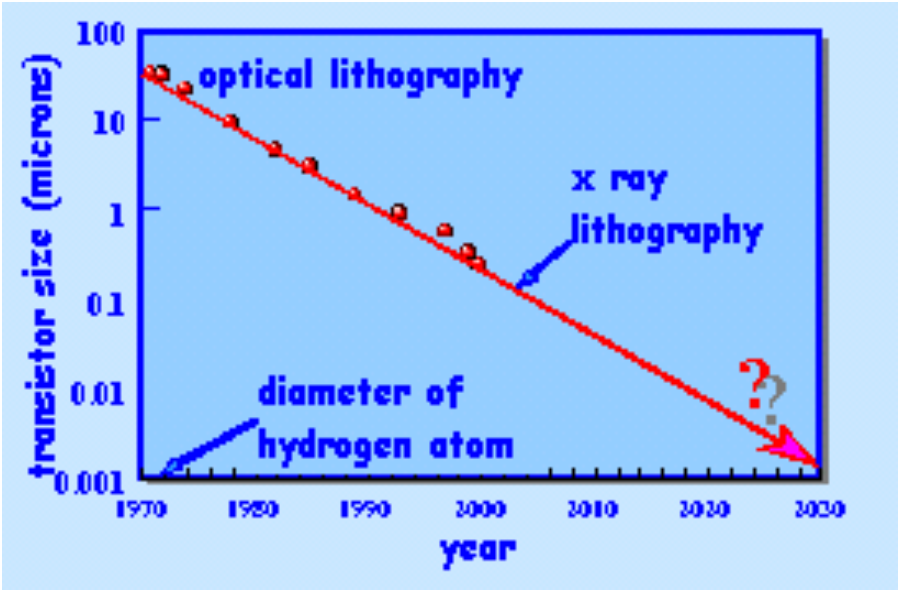


The Trash 80

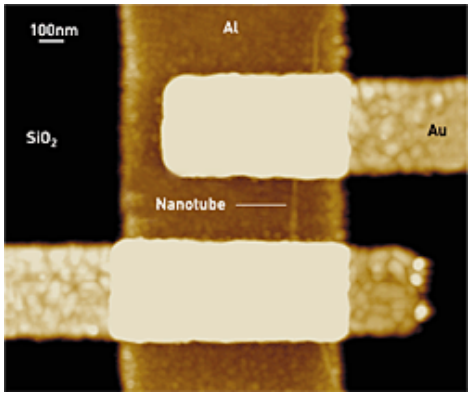


What happens when you try to simulate orbits but you don't know trigonometry (circa 1984)

# Quantum Machines

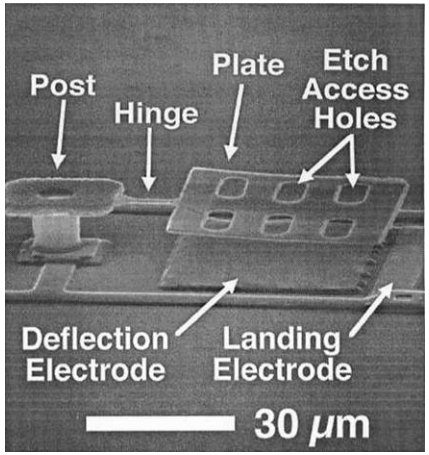



Moore's Law




Dekker, Delft University

Molecular Transistors




ICL, Stanford University

MEMS

The coming of the age of quantum machines.

# Outline

- Introduction to Quantum Error Correction
- Stabilizer Codes
- Quantum Codes in Physical Systems
- The Future

# A Critical Ghost



All papers on quantum computing should carry a footnote: *"This proposal, like all proposals for quantum computation, relies on speculative technology, does not in its current form take into account all possible sources of noise, unreliability and manufacturing error, and probably will not work."*

**Rolf Landauer**
**IBM**

Get around Landauer's objections by finding systems which naturally enact quantum error correction.

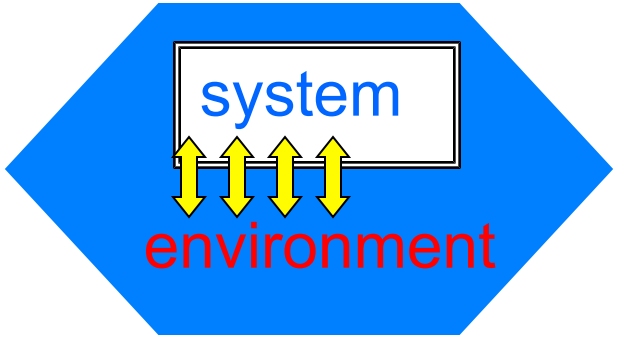


**Nature abhors a quantum computer?**

# The Problem

T/F? The evolution of a quantum system is always unitary.




$\mathbf{H}_S$

Hamiltonian

$$\mathbf{H}_{SE} = \mathbf{H}_S \otimes \mathbf{I}_B + \mathbf{I}_S \otimes \mathbf{H}_E + \mathbf{H}_I$$

$$\mathbf{U}(t) = \exp\left[-i\mathbf{H}_S t\right]$$

Evolution

$$\mathbf{U}(t) = \exp\left[-i\mathbf{H}_{SE} t\right]$$

$$\boldsymbol{\rho}(t) = \mathbf{U}(t)\boldsymbol{\rho}(0)\mathbf{U}^\dagger(t)$$

$$\boldsymbol{\rho}(t) = \sum_k \mathbf{A}_k(t)\boldsymbol{\rho}(0)\mathbf{A}_k^\dagger(t)$$

$$\mathbf{U}^\dagger(t)\mathbf{U}(t) = \mathbf{I}$$

$$\sum_k \mathbf{A}_k^\dagger(t)\mathbf{A}_k(t) = \mathbf{I}$$

# Evolution

Paulis :

$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Operator Sum Representation (a.k.a. Superoperator)

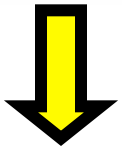
$$\boldsymbol{\rho}(t) = \sum_k \mathbf{A}_k(t)\boldsymbol{\rho}(0)\mathbf{A}_k^\dagger(t) \qquad \sum_k \mathbf{A}_k^\dagger(t)\mathbf{A}_k(t) = \mathbf{I}$$

Example: dephasing channel

$$\mathbf{A}_0 = \sqrt{1-p}\mathbf{I}$$
$$\mathbf{A}_1 = \sqrt{p}\mathbf{Z}$$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

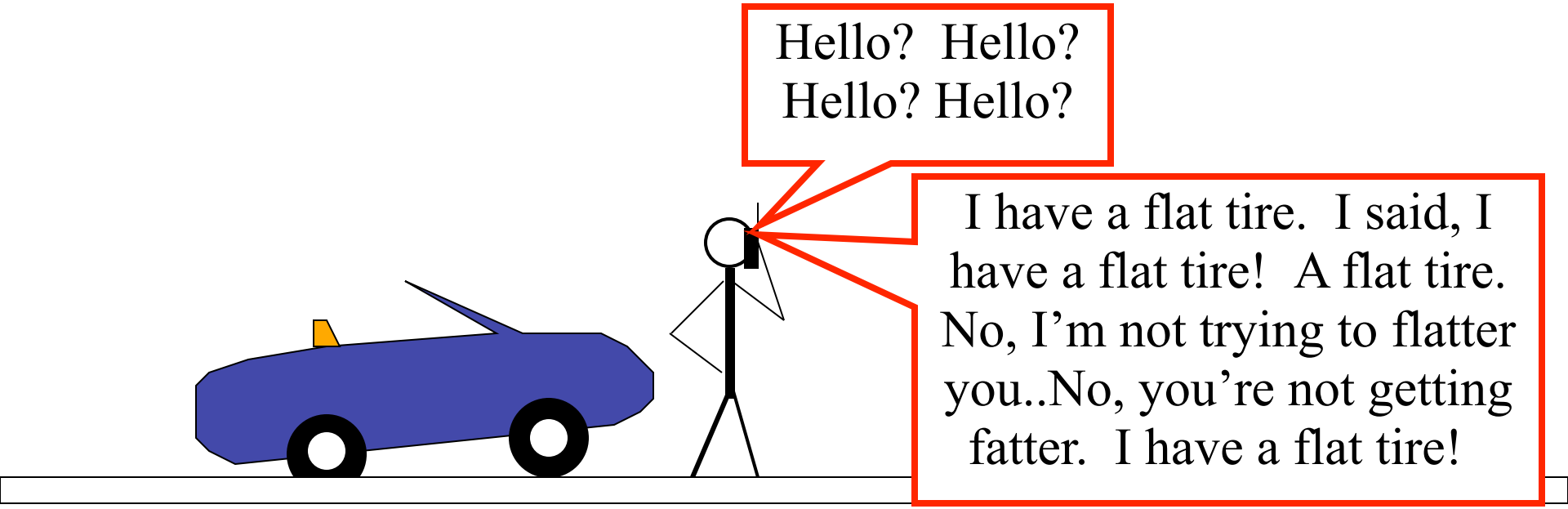
$$\boldsymbol{\rho} = |\psi\rangle\langle\psi| = \begin{bmatrix} |\alpha|^2 & \alpha\beta^* \\ \alpha^*\beta & |\beta|^2 \end{bmatrix}$$



decay of off-diagonal elements

$$(1-p)\boldsymbol{\rho} + p\mathbf{Z}\boldsymbol{\rho}\mathbf{Z} = \begin{bmatrix} |\alpha|^2 & (1-2p)\alpha\beta^* \\ (1-2p)\alpha^*\beta & |\beta|^2 \end{bmatrix}$$

# Noisy Cell Phone

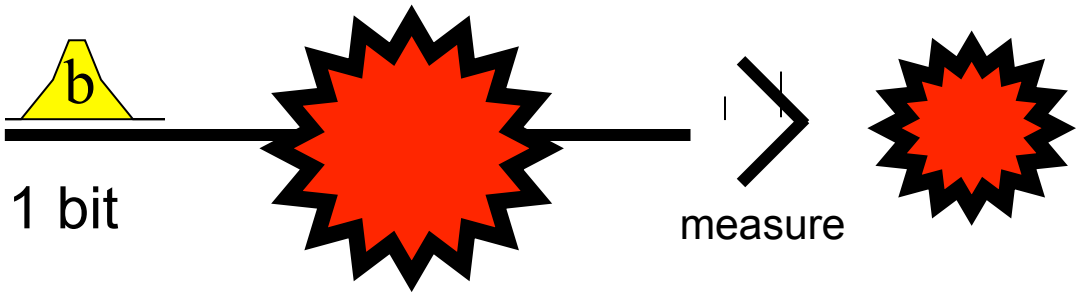


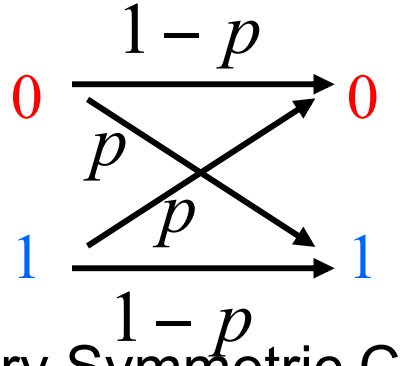

Communication over a noisy CHANNEL can be overcome via

**ENCODING**

"Hello?" = "Hello? Hello? Hello? Hello?" [using redundancy to encode "Hello"]

# Classical Error Correction




Binary Symmetric Channel

Probability of error $= p$

Encode:

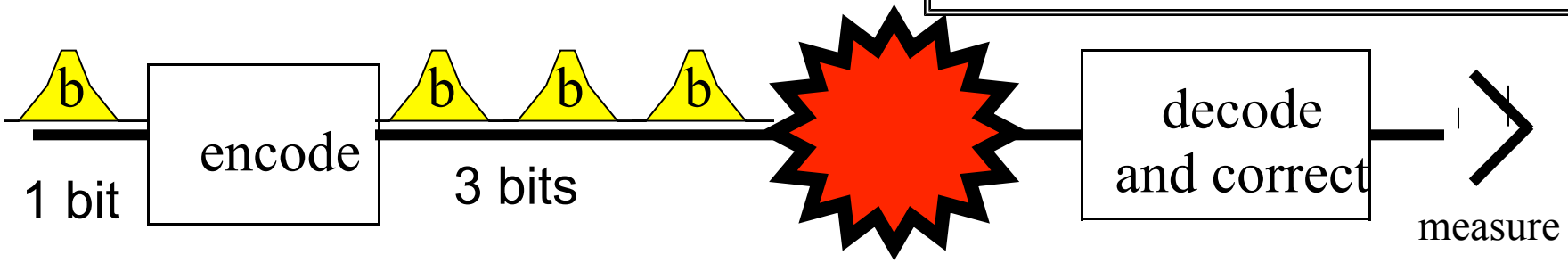
$0 \rightarrow 000$

$1 \rightarrow 111$

Decode and correct:

000, 001, 010, 100 → 0

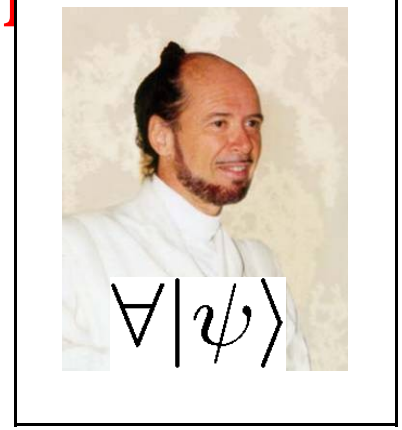
111, 110, 101, 011 → 1




Probability of error $= 3p^2(1-p) + p^3 =< p$ if $p < \frac{1}{2}$

# Reason to be a Pessimist

*No cloning*:

$|\psi\rangle$ →  → $|\psi\rangle \otimes |\psi\rangle$



Quantum Cloning Machine

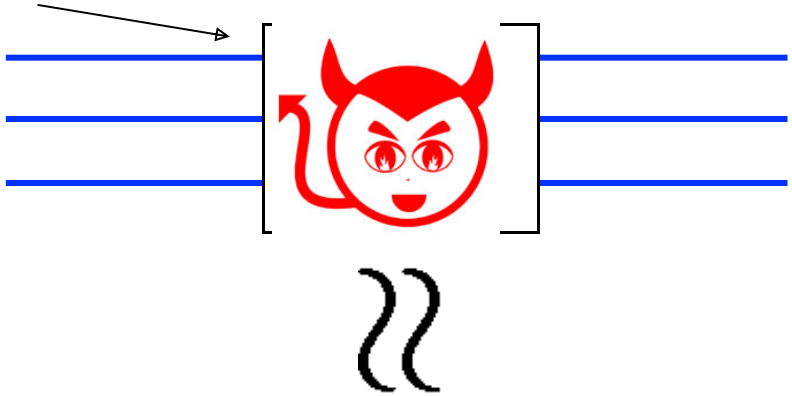
"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

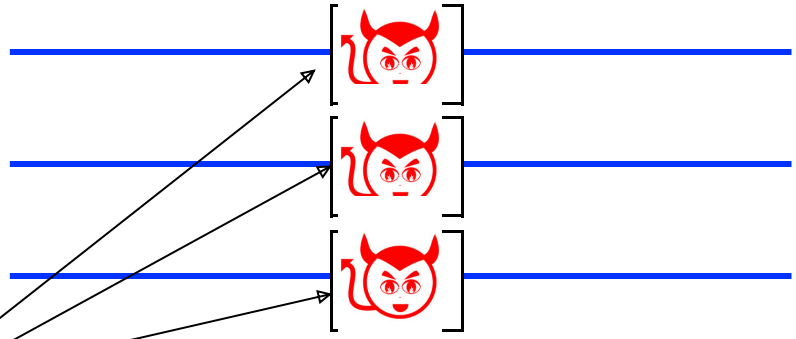
No quantum repetition code:

$$|\psi\rangle \nrightarrow |\psi\rangle \otimes |\psi\rangle \otimes |\psi\rangle$$

# Unrealistic Realistic Channel

$$\mathcal{S} = \{\mathrm{I}, \mathrm{X}_1, \mathrm{X}_2, \mathrm{X}_3\},\ p_1 = 1 - 3\epsilon, p_2 = p_3 = p_4 = \epsilon$$





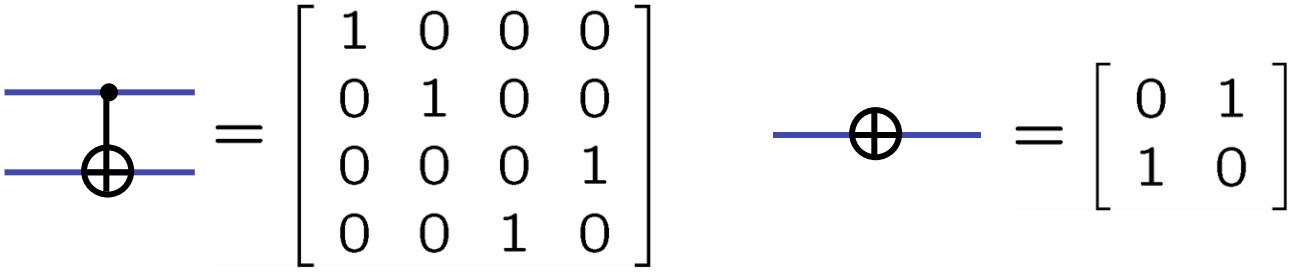
$$\mathcal{S} = \{\mathrm{I}, \mathrm{X}\},\ p_1 = 1 - \epsilon, p_2 = \epsilon$$

notation

$$\mathrm{P}_i = \mathrm{I} \otimes \cdots \otimes \mathrm{I} \otimes \underbrace{\mathrm{P}}_{ith\ \ qubit} \otimes \mathrm{I} \otimes \cdots \otimes \mathrm{I}$$

# Baby Steps

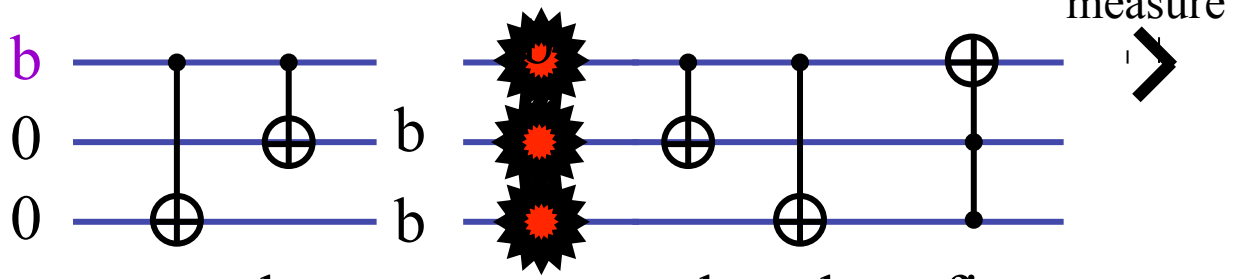




$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

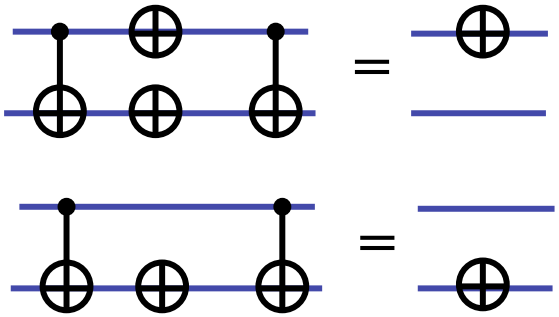
WWCCD? (What Would Classical Coders Do?)

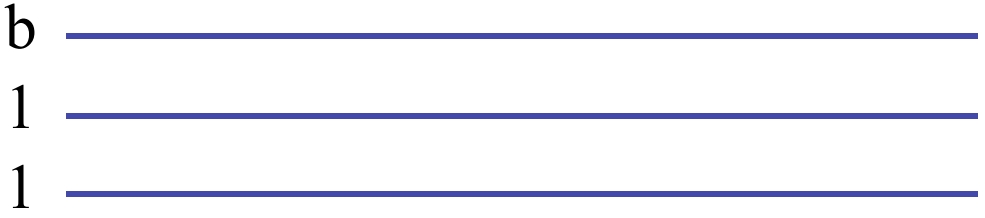
0 □000, 1 □ 111




100 □ 111□101 □ 110 □ 110

identities



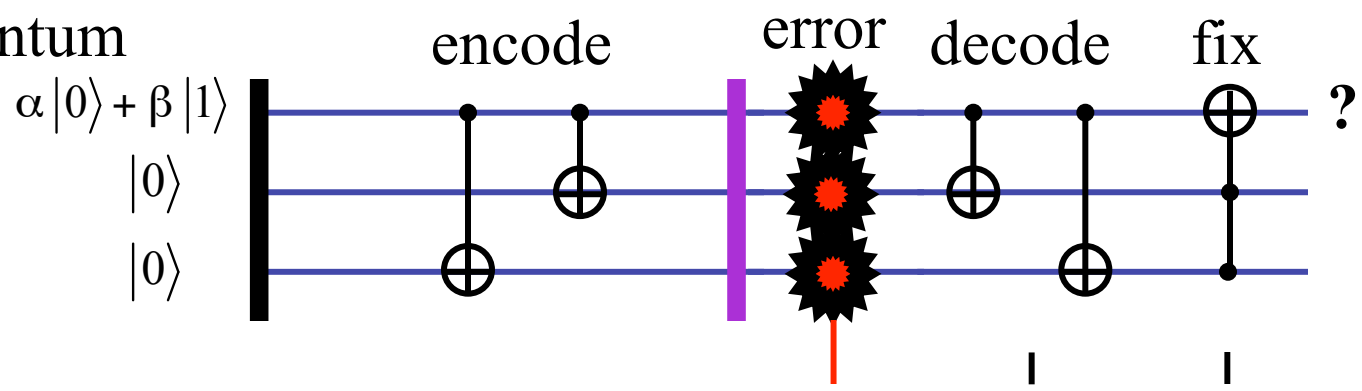



error #@%

# Naïve

4. operator identities still hold

Lets be naïve, take classical and move over to quantum




$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|0\rangle \otimes |0\rangle \otimes |0\rangle + \beta|1\rangle \otimes |1\rangle \otimes |1\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{XII}{\rightarrow} \alpha|100\rangle + \beta|011\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|11\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IXI}{\rightarrow} \alpha|010\rangle + \beta|101\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|10\rangle$$

$$\alpha|000\rangle + \beta|111\rangle \overset{IIX}{\rightarrow} \alpha|001\rangle + \beta|110\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)|01\rangle$$

error decode fix

encoded into subspace:

$|0\rangle \rightarrow |000\rangle, |1\rangle \rightarrow |111\rangle$

(no-cloning evaded!)

2. errors take to orthogonal subspaces + maintain orthogonality

3. syndrome

$\mathbf{XII}|000\rangle = |100\rangle, \mathbf{XII}|111\rangle = |011\rangle$

# OK Wise Guy

What about "phase" errors? $\rho_{out} = (1-p)\rho_{in} + p\mathbf{Z}\rho_{in}\mathbf{Z}$

phase error: $\mathbf{Z}(\alpha|0\rangle + \beta|1\rangle) = (\alpha|0\rangle - \beta|1\rangle)$ …sort of not classical error

$\mathcal{S} = \{\mathbf{I}, \mathbf{Z}_1, \mathbf{Z}_2, \mathbf{Z}_3\},\ p_1 = 1-3\epsilon, p_2 = p_3 = p_4 = \epsilon$

Wise guy says "basis change please":

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$$

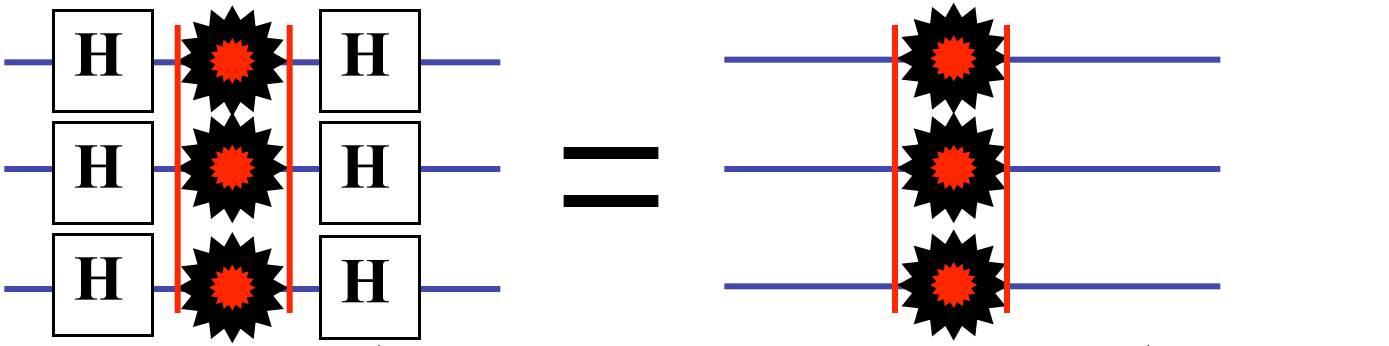
$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$\mathbf{Z}|+\rangle = |-\rangle$$

$$\mathbf{Z}|-\rangle = |+\rangle$$

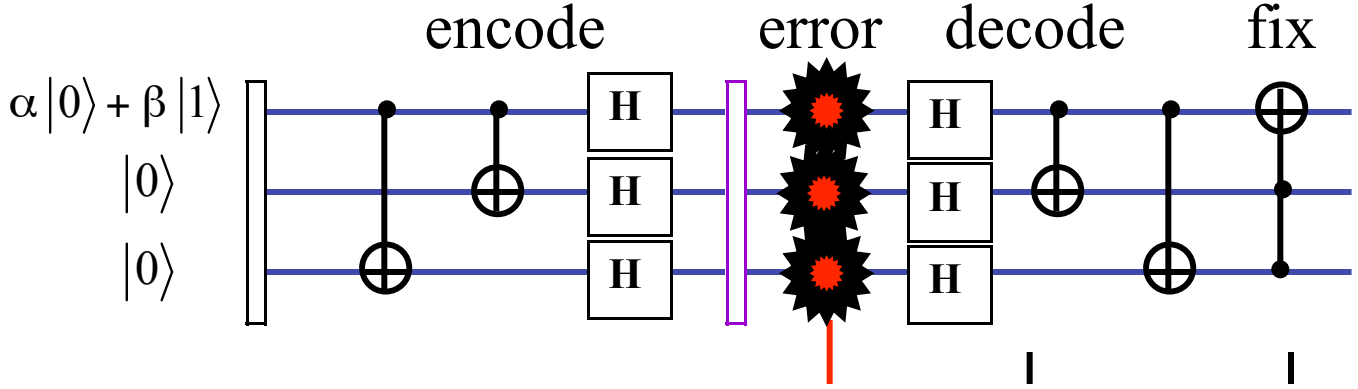
looks like bit flip error in this new basis!

$$\mathbf{HZH} = \mathbf{X}, \quad \mathbf{H} = |+\rangle\langle 0| + |-\rangle\langle 1|$$




phase errors $= \mathbf{Z}$ bit flip errors $= \mathbf{X}$

# Molly: "I love you, I really love you"
# Sam: "Ditto."




$$(\alpha|0\rangle + \beta|1\rangle) \otimes |0\rangle \otimes |0\rangle \xrightarrow{U} \alpha|+\rangle \otimes |+\rangle \otimes |+\rangle + \beta|-\rangle \otimes |-\rangle \otimes |-\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{ZII}{\rightarrow} \alpha|-++\rangle + \beta|+--\rangle \rightarrow (\alpha|1\rangle + \beta|0\rangle)\,|11\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|11\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IZI}{\rightarrow} \alpha|+-+\rangle + \beta|-+-\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|10\rangle$$

$$\alpha|+++\rangle + \beta|---\rangle \overset{IIZ}{\rightarrow} \alpha|++-\rangle + \beta|--+\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle \rightarrow (\alpha|0\rangle + \beta|1\rangle)\,|01\rangle$$

error decode fix

<u>encoded into subspace</u>:

$|0\rangle \rightarrow |+++\rangle, |1\rangle \rightarrow |---\rangle$

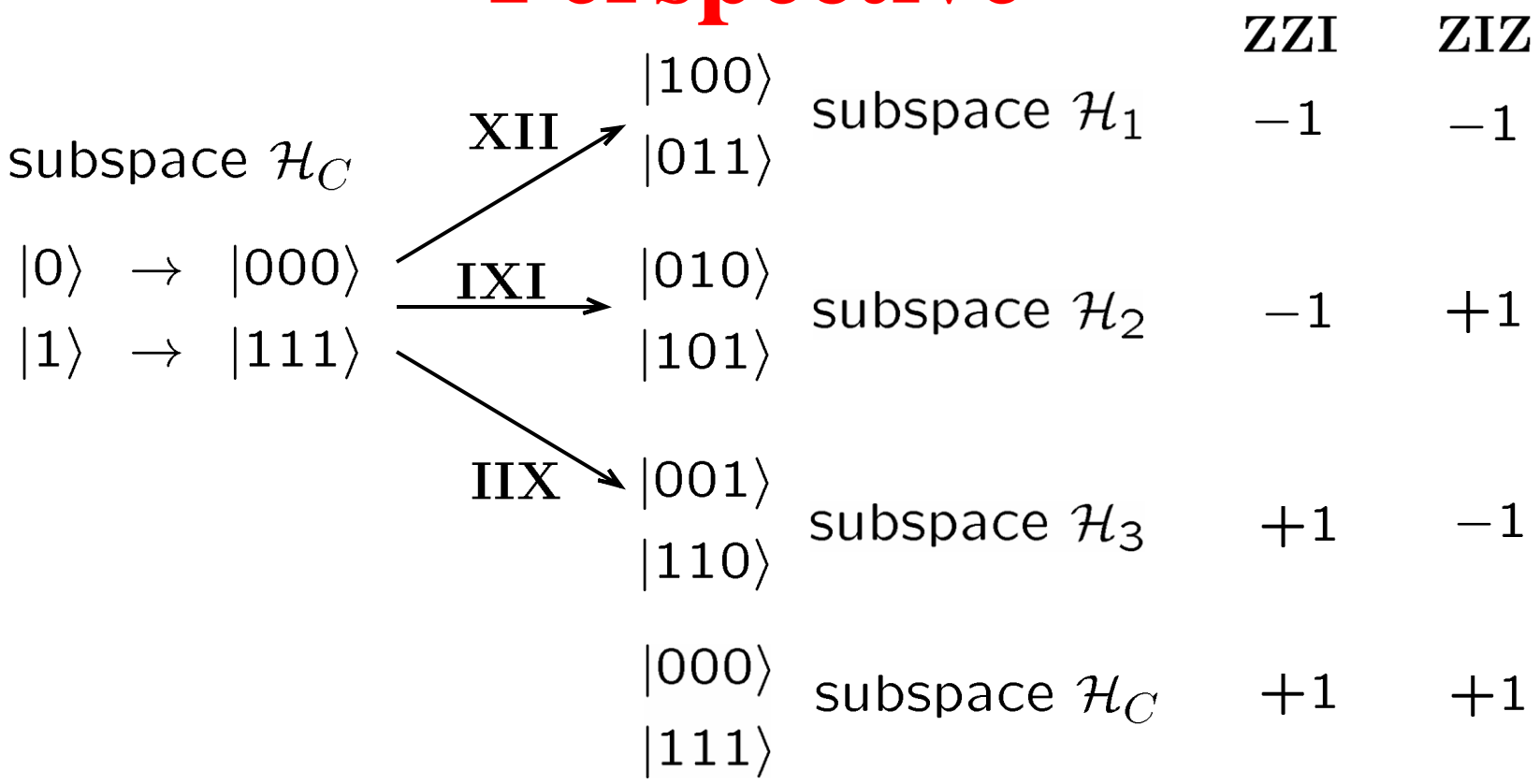
(no-cloning evaded!)

2. <u>errors take to orthogonal subspaces + maintain orthogonality</u>

$\mathbf{ZII}|+++\rangle = |-++\rangle, \mathbf{ZII}|---\rangle = |+--\rangle$

3. <u>syndrome</u>

# Perspective




subspaces $\mathcal{H}_C$, $\mathcal{H}_1$, $\mathcal{H}_2$, and $\mathcal{H}_3$ are orthogonal

$\dim \mathcal{H}_C = \dim \mathcal{H}_1 = \dim \mathcal{H}_2 = \dim \mathcal{H}_3 = 2$

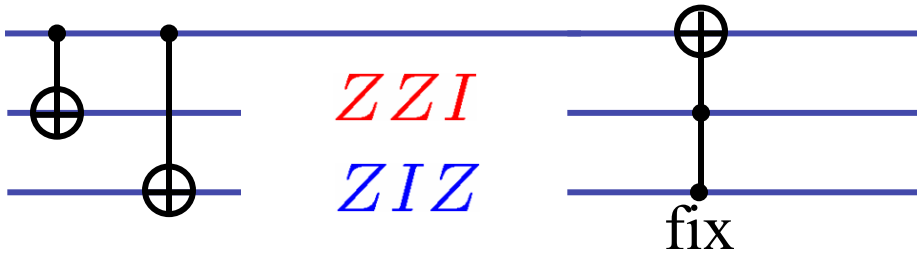
Orthogonal subspaces can be distinguished by measurements

without measuring information encoded into the subspace!!!

# Not An Optical Illusion




| | | $b$ | ZZ |
|---|---|---|---|
| $\|00\rangle$ | $\rightarrow$ | $\|00\rangle$ | $+1$ |
| $\|01\rangle$ | $\rightarrow$ | $\|01\rangle$ | $-1$ |
| $\|10\rangle$ | $\rightarrow$ | $\|11\rangle$ | $-1$ |
| $\|11\rangle$ | $\rightarrow$ | $\|10\rangle$ | $+1$ |




| | | | | | |
|---|---|---|---|---|---|
| | | XII | $\|100\rangle$ | $\|111\rangle$ | $\|011\rangle$ |
| | | | $\|011\rangle$ | $\|011\rangle$ | $\|111\rangle$ |
| $\|0\rangle$ | $\rightarrow$ $\|000\rangle$ | IXI | $\|010\rangle$ | $\|010\rangle$ | $\|010\rangle$ |
| $\|1\rangle$ | $\rightarrow$ $\|111\rangle$ | | $\|101\rangle$ | $\|110\rangle$ | $\|110\rangle$ |
| | | IIX | $\|001\rangle$ | $\|001\rangle$ | $\|001\rangle$ |
| | | error | $\|110\rangle$ | $\|101\rangle$ | $\|101\rangle$ |

# Encoding Away Your Ills

| 3 qubit bit flip code | 3 qubit phase flip code |
|---|---|
| $\|0\rangle \rightarrow \|000\rangle$ | $\|0\rangle \rightarrow \|+++\rangle$ |
| $\|1\rangle \rightarrow \|111\rangle$ | $\|1\rangle \rightarrow \|---\rangle$ |

phase errors **ZII**, **IZI**, **IIZ** act as ■ on bit flip code qubits:

$$|0\rangle_B = |000\rangle \qquad \mathbf{ZII}|0\rangle_B = |0\rangle_B$$
$$|1\rangle_B = |111\rangle \qquad \mathbf{ZII}|1\rangle_B = -|1\rangle_B$$

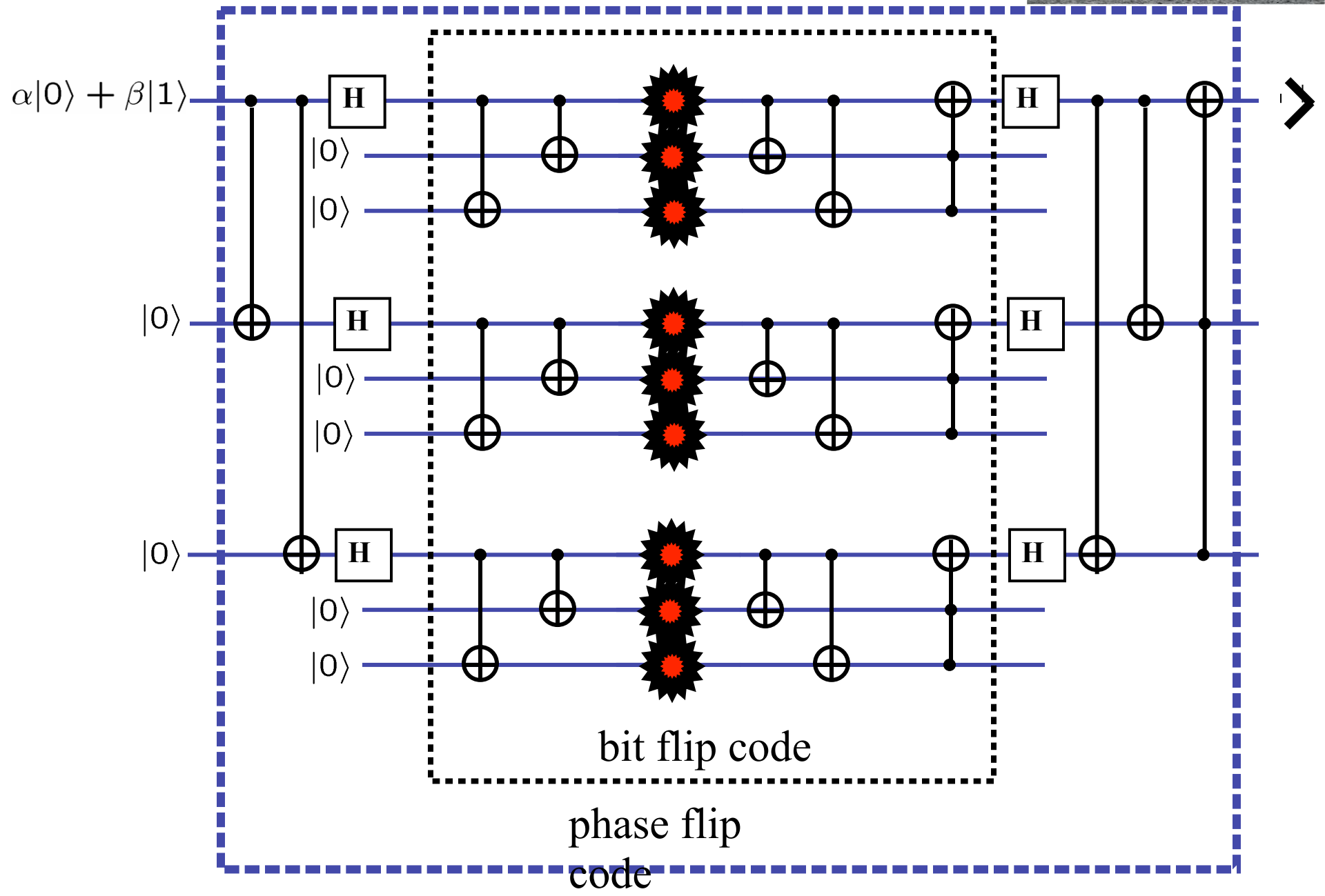
define:
$$|p\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B + |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$$
$$|m\rangle = \frac{1}{\sqrt{2}}(|0\rangle_B - |1\rangle_B) = \frac{1}{\sqrt{2}}(|000\rangle - |111\rangle)$$

Shor Code: (Peter Shor, 1995)

$$|0\rangle \rightarrow |ppp\rangle = \frac{1}{2\sqrt{2}}(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)(|000\rangle + |111\rangle)$$
$$|1\rangle \rightarrow |mmm\rangle = \frac{1}{2\sqrt{2}}(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)(|000\rangle - |111\rangle)$$

# Inside Shor




$\alpha|0\rangle + \beta|1\rangle$
$|0\rangle$
$|0\rangle$
H
H
H
H
H
H
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
$|0\rangle$
bit flip code
phase flip
code

# Lets Get Definitional

A quantum code $C$ is simply a subspace.

## Quantum Error Correcting Code

A quantum error correcting code $C$ for the quantum operation $\mathcal{E}$ is a code for which there exists a recovery operator $\mathcal{R}$ such that $\mathcal{R} \circ \mathcal{E}$ acts proportional to identity on the codespace.

## Iff Condition for Quantum Error Correcting Code

A code $C$ with orthonormal basis $\{|i\rangle\}$ is a quantum error correcting code for the quantum operation $\mathcal{E}$ with operation elements $\{\mathbf{E}_k\}$ iff

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

# Linear Combination of Errors

Quantum Error Correcting Code

A quantum error correcting code $C$ for the quantum operation $\mathcal{E}$ is a code for which there exists a recovery operator $\mathcal{R}$ such that $\mathcal{R} \circ \mathcal{E}$ acts proportional to identity on the codespace.

If

$$\sum_l \mathbf{R}_k \left[ \sum_k \mathbf{A}_k \boldsymbol{\rho} \mathbf{A}_k^\dagger \right] \mathbf{R}_k^\dagger = \boldsymbol{\rho}$$

then

$$\sum_l \mathbf{R}_k \left[ \sum_k \mathbf{B}_k \boldsymbol{\rho} \mathbf{B}_k^\dagger \right] \mathbf{R}_k^\dagger = \boldsymbol{\rho} \qquad \mathbf{B}_k = \sum_i b_{ki} \mathbf{A}_i$$

This allows us to digitize the errors

# In Action

$$\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = C_{ab}\delta_{ij}$$

$|0_L\rangle = |000\rangle$ error set $\mathbf{E}_a \in \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3\}$

$|1_L\rangle = |111\rangle$ $\mathbf{E}_b^\dagger \mathbf{E}_a \in \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{X}_1\mathbf{X}_2, \mathbf{X}_1\mathbf{X}_3, \mathbf{X}_2\mathbf{X}_3\}$

If $\mathbf{E}_b^\dagger \mathbf{E}_a \neq \mathbf{I}$, $\langle i|\mathbf{E}_b^\dagger \mathbf{E}_a|j\rangle = 0$

---

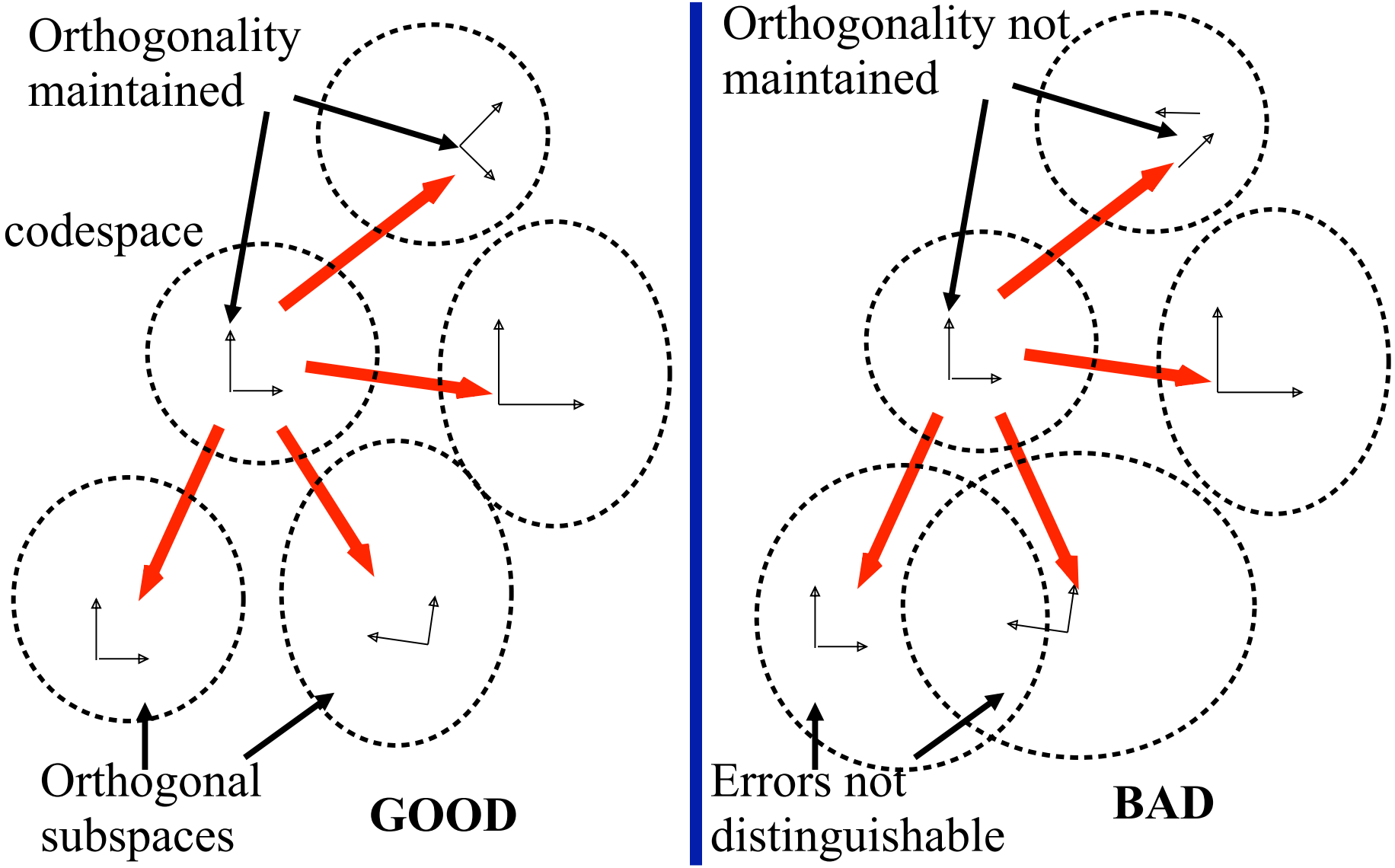
$|0_L\rangle = |000\rangle$ error set $\mathbf{E}_a \in \{\mathbf{I}, \mathbf{X}_1, \mathbf{X}_2, \mathbf{X}_3, \mathbf{Z}_1\}$

$|1_L\rangle = |111\rangle$

$\mathbf{E}_b^\dagger = \mathbf{I}$, $\mathbf{E}_a = \mathbf{Z}_1$, $\langle 0_L|\mathbf{Z}_1|0_L\rangle = 1$ but $\langle 1_L|\mathbf{Z}_1|1_L\rangle = -1$

# Pictures

$$\langle i|\mathbf{F}_a^\dagger \mathbf{F}_b|j\rangle = D_a \delta_{ab}\delta_{ij}$$



# A Code

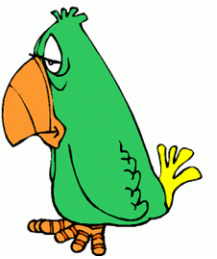
A five qubit code which protects against one quantum error:

$$|0\rangle \rightarrow \frac{1}{4}[-|00000\rangle - |11000\rangle - |01100\rangle - |00110\rangle - |00011\rangle - |10001\rangle + |10010\rangle + |10100\rangle + |01001\rangle + |01010\rangle + |00101\rangle + |11110\rangle + |11101\rangle + |11011\rangle + |10111\rangle + |01111\rangle]$$

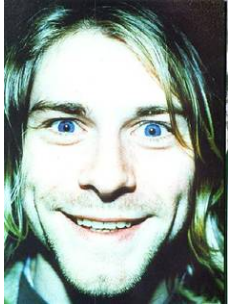
$$|1\rangle \rightarrow \frac{1}{4}[-|11111\rangle - |00111\rangle - |10011\rangle - |11001\rangle - |11100\rangle - |01110\rangle + |01101\rangle + |01011\rangle + |10110\rangle + |10101\rangle + |11010\rangle + |00001\rangle + |00010\rangle + |00100\rangle + |01000\rangle + |10000\rangle]$$

Better way to talk about the code than the states:

**the Stabilizer formalism**



# Pauli Want a Cracker?



$$\mathbf{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix} \quad \mathbf{Z} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\{\mathbf{X}, \mathbf{Y}\} = \{\mathbf{X}, \mathbf{Z}\} = \{\mathbf{Y}, \mathbf{Z}\} = 0$$

**def:** Pauli group on 1 qubit: $\mathcal{P}_1 = \{\pm\mathbf{I}, \pm i\mathbf{I}, \pm\mathbf{X}, \pm i\mathbf{X}, \pm\mathbf{Y}, \pm i\mathbf{Y}, \pm\mathbf{Z}, \pm i\mathbf{Z}\}$

**def:** Pauli operator on $n$ qubits: $i^k \mathbf{P}_1 \mathbf{P}_2 \cdots \mathbf{P}_n$

$\mathbf{P}_i \in \mathcal{P}_1,\ k = 0 \ldots 3$

**def:** Pauli group on $n$ qubits: $\mathcal{P}_n = \{\text{all Pauli operators on } n \text{ qubits}\}$

# Pauli Group Facts

Any elements $\mathbf{P}, \mathbf{Q} \in \mathcal{P}_n$ either commute or anticommute, but not both

$$[\mathbf{P}, \mathbf{Q}] = 0 \quad \text{or} \quad \{\mathbf{P}, \mathbf{Q}\} = 0$$

$$\begin{aligned}\mathbf{PQ} &= \mathbf{P}_1\mathbf{P}_2\cdots\mathbf{P}_n\mathbf{Q}_1\mathbf{Q}_2\cdots\mathbf{Q}_n = (\mathbf{P}_1\mathbf{Q}_1)(\mathbf{P}_2\mathbf{Q}_2)\cdots(\mathbf{P}_n\mathbf{Q}_n) \\ &= (\pm\mathbf{Q}_1\mathbf{P}_1)(\pm\mathbf{Q}_2\mathbf{P}_2)\cdots(\pm\mathbf{Q}_n\mathbf{P}_n) = \pm\mathbf{QP}\end{aligned}$$

$\mathbf{X}_1\mathbf{X}_2\mathbf{I}_3\mathbf{Z}_4$

$\mathbf{X}_1\mathbf{Z}_2\mathbf{Z}_3\mathbf{Y}_4$ $\qquad (\mathbf{X}_1\mathbf{X}_2\mathbf{I}_3\mathbf{Z}_4)(\mathbf{X}_1\mathbf{Z}_2\mathbf{Z}_3\mathbf{Y}_4) = (\mathbf{X}_1\mathbf{X}_2\mathbf{I}_3\mathbf{Z}_4)(\mathbf{X}_1\mathbf{Z}_2\mathbf{Z}_3\mathbf{Y}_4)$

$\downarrow\ \downarrow\ \downarrow\ \downarrow$

$s\ d\ t\ d$

$s =$ same, $t =$ one is trivial, $d =$ different (none trivial)

$d$ is even $\Rightarrow \mathbf{PQ} = \mathbf{QP} \Rightarrow [\mathbf{P}, \mathbf{Q}] = 0$

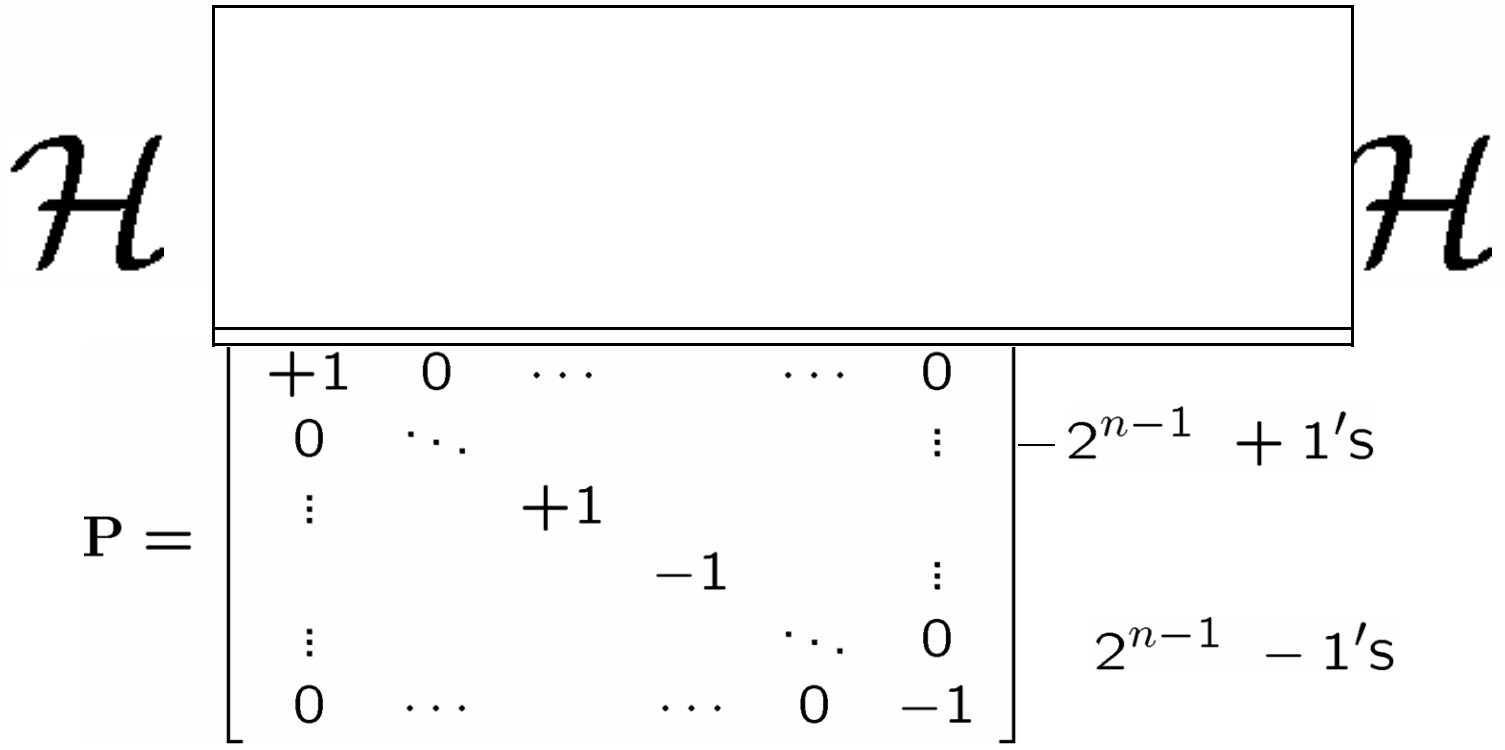
$d$ is odd $\Rightarrow \mathbf{PQ} = -\mathbf{QP} \Rightarrow \{\mathbf{P}, \mathbf{Q}\} = 0$

Pauli elements, nonidentity are traceless $\mathrm{Tr}\,[\mathbf{P}] = 0$, $\mathbf{P} \neq i^k\mathbf{I}$ and all elements square to $\pm$ identity $\mathbf{P}^2 = \pm\mathbf{I}$

# Set Your Square Phasors to +1

Focus on the elements of $\mathtt{P}_n$ such that $\mathbf{P}^2 = \mathbf{I}$:

- Eigenvalues of such $\mathbf{P}$ are $\pm 1$: $\mathbf{P}^2|\psi\rangle = \lambda^2|\psi\rangle = |\psi\rangle \rightarrow \lambda = \pm 1$.
- Degeneracies of $+1$ and $-1$ eigenvalues are $2^{n-1}$ for nonidentity $\mathbf{P}$:




The $\mathbf{P}$ divide the space in half depending on the $\pm 1$ eigenvalue.

# Abelian not Reptilian

A *Pauli Stabilizer Group* on $n$ qubits is an Abelian Subgroup of the Pauli Group on $n$ qubits whose elements all square to identity.

$$\mathrm{S} = \{\mathbf{S}_\alpha \in \mathrm{P} | \mathbf{S}_\alpha^2 = \mathbf{I} \text{ and } [\mathbf{S}_\alpha, \mathbf{S}_\beta] = 0, \forall \alpha, \beta\}$$

Example: $\mathrm{S}_x = \{\mathbf{IIII}, \mathbf{XXXX}, \mathbf{YYYY}, \mathbf{ZZZZ}\}$

The *generators* of a Pauli Stabilizer Group $\mathrm{S}$ is the smallest set whose elements can be multiplied to obtain the entire group [not unique].

Example: $\mathrm{S}_x$ is generated by $\{\mathbf{XXXX}, \mathbf{ZZZZ}\}$

$$\mathrm{S}_x = \langle \mathbf{XXXX}, \mathbf{ZZZZ} \rangle$$

Since group is abelian and square to identity, every element can be expressed as

$$\mathbf{S}_1^{\alpha_1} \mathbf{S}_2^{\alpha_2} \cdots \mathbf{S}_l^{\alpha_l}$$

where $\mathbf{S}_i$ generate the group and $\alpha_i \in \{0, 1\}$.

Example: $\mathbf{S} = (\mathbf{XXXX})^{\alpha_1} (\mathbf{ZZZZ})^{\alpha_2}, \ \forall \ \mathbf{S} \in \mathrm{S}_x$



# Pauli Stabilizer Code

A Pauli Stabilizer Code $S$ for a Pauli Stabilizer Group $\mathbb{S}$ is defined as the common $+1$ eigenvalue eigenspace of the elements of $\mathbb{S}$:

$$|\psi_C\rangle \in S \text{ iff } \mathbf{S}_\alpha|\psi_C\rangle = |\psi_C\rangle, \ \forall \mathbf{S}_\alpha \in \mathbb{S}$$

Since

$$\mathbf{S}(\alpha_1, \alpha_2, \ldots, \alpha_k) = \mathbf{S}_1^{\alpha_1}\mathbf{S}_2^{\alpha_2}\cdots\mathbf{S}_k^{\alpha_k}$$

we need only check generators:

$$|\psi_C\rangle \in S \text{ iff } \mathbf{S}_\alpha|\psi_C\rangle = |\psi_C\rangle, \ \forall \mathbf{S}_\alpha \in \langle\mathbb{S}\rangle$$

# Truth By Example

Example: $\mathrm{S}_x = \langle \mathbf{XXXX}, \mathbf{ZZZZ}, \mathbf{XXII} \rangle$

$\mathbf{ZZZZ}|\psi_C\rangle = |\psi_C\rangle \Rightarrow$ computational basis states are even parity

$$|\psi_C\rangle \in \mathrm{Span}\,[|0000\rangle, |0011\rangle, |0110\rangle, |0101\rangle, |1100\rangle, |1010\rangle, |1001\rangle, |1111\rangle]$$

$\mathbf{XXXX}|\psi_C\rangle = |\psi_C\rangle \Rightarrow$ use projector $\frac{1}{2}(\mathbf{IIII} + \mathbf{XXXX})$

$$|\psi_C\rangle \in \mathrm{Span}\left[\frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle), \frac{1}{\sqrt{2}}(|0110\rangle + |1001\rangle), \frac{1}{\sqrt{2}}(|1100\rangle + |0011\rangle), \frac{1}{\sqrt{2}}(|1010\rangle + |0101\rangle)\right]$$

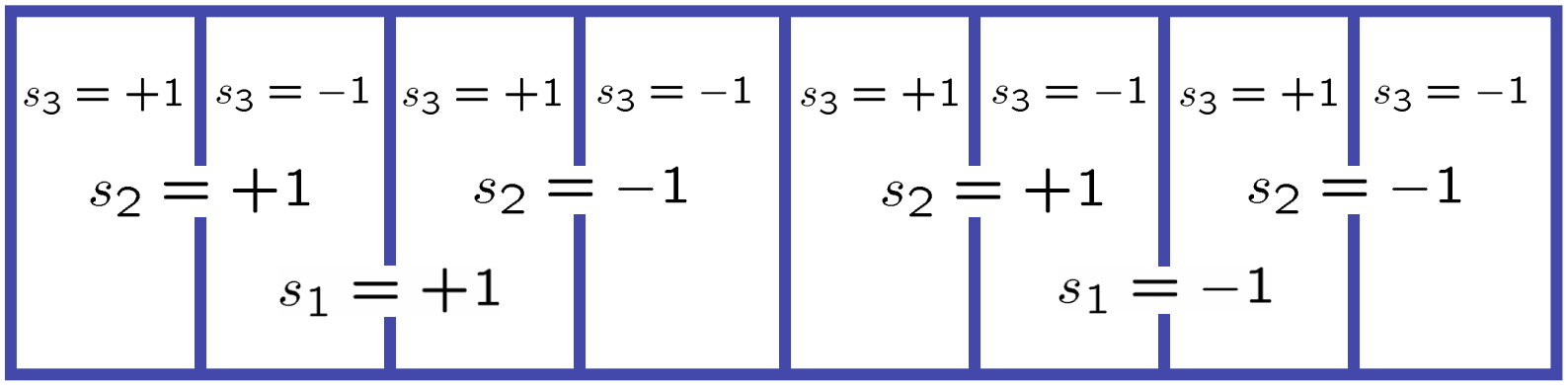
$\mathbf{XXII}|\psi_C\rangle = |\psi_C\rangle \Rightarrow$ use projector $\frac{1}{2}(\mathbf{IIII} + \mathbf{XXII})$

$$|\psi_C\rangle \in \mathrm{Span}\left[\frac{1}{2}(|0000\rangle + |1111\rangle + |1100\rangle + |0011\rangle), \frac{1}{2}(|0110\rangle + |0110\rangle + |1010\rangle + |0101\rangle)\right]$$

# Stabilizer Subspaces

$\mathbf{G}_i|\psi\rangle = |\psi\rangle$, $\forall i$ defines a $2^{n-l}$ dimensional subspace.

Every unique $s_i = \pm 1$ defines, via $\mathbf{G}_i|\psi\rangle = s_i|\psi\rangle$, $\forall i$, a *unique* subspace.




$2^n$ dimensional Hilbert space

| $s_1$ | $s_2$ | |
|---|---|---|
| $+1$ | $+1$ | $2^{n-2}$ dimensional codespace |
| $+1$ | $-1$ | $2^{n-2}$ dimensional codespace |
| $-1$ | $+1$ | $2^{n-2}$ dimensional codespace |
| $-1$ | $-1$ | $2^{n-2}$ dimensional codespace |

$|s_1, s_2, \ldots, s_k, m\rangle$, $m$ labels $2^{n-l}$ subspace

# Stabilizer Nerror

Stabilizer $\mathbb{S}$ with generators $\mathbf{G}_1, \ldots, \mathbf{G}_k$ stabilizes a code of $n-l$ qubits. What errors does this code correct?

Suppose $\mathbf{E}$ anticommutes with *at least* one $\mathbf{G}_i$,

$$\{\mathbf{E}, \mathbf{G}_i\} = 0 \quad \mathbf{E}\mathbf{G}_i = -\mathbf{G}_i\mathbf{E}$$

then as before, for stabilized codewords,

$$\langle i|\mathbf{E}|j\rangle = \langle i|\mathbf{E}\mathbf{G}_i|j\rangle = -\langle i|\mathbf{G}_i\mathbf{E}|j\rangle = -\langle i|\mathbf{G}|j\rangle = 0$$

If $\mathbf{E}$ is in the stabilizer:

$$\langle i|\mathbf{E}|j\rangle = \langle i|j\rangle = \delta_{ij}$$

Given a set of errors $\mathbf{E}_a$, a stabilizer code $S$ corrects these errors if the products $\mathbf{E}_a^\dagger \mathbf{E}_b$ either

(i) anticommute with at least one stabilizer generator

(ii) are in the stabilzer group

# Surfing The Subspaces

If $\mathbf{E}_a$ anticommutes with $\mathbf{G}_i$,

$$\mathbf{G}_i(\mathbf{E}_a|s_1,\ldots,s_l,m\rangle) = -s_i(\mathbf{E}_a|s_1,\ldots,s_l,m\rangle)$$

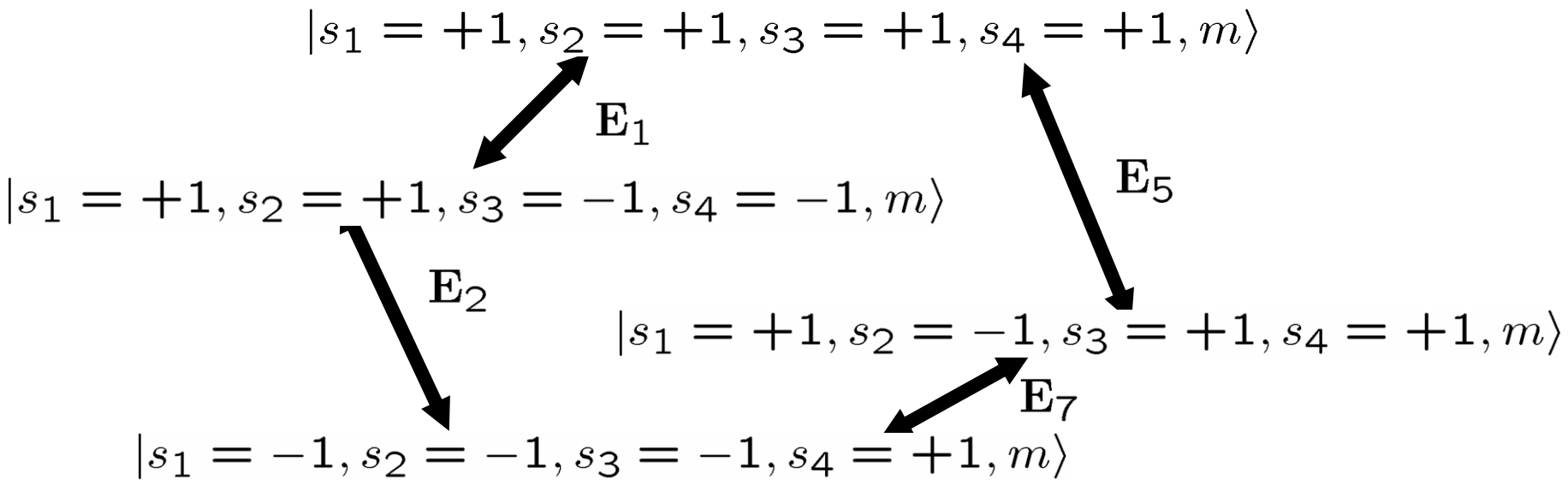
$$|s_1,s_2,s_3,\ldots,s_l,m\rangle$$

$\mathbf{E}_a$

$$|\bar{s}_1,\bar{s}_2,\bar{s}_3,\ldots,\bar{s}_l,m\rangle$$

$\bar{s}_i = s_i$ if $[\mathbf{G}_i,\mathbf{E}_a] = 0$

$\bar{s}_i = -s_i$ if $\{\mathbf{G}_i,\mathbf{E}_a\} = 0$



# Examples

$\langle \mathbb{S} \rangle = \{\mathbf{ZZI}, \mathbf{IZZ}\}$ corrects single bit flip errors $\{\mathbf{E}_a\} = \{\mathbf{XII}, \mathbf{IXI}, \mathbf{IIX}\}$.

$\langle \mathbb{S} \rangle = \{\mathbf{XXI}, \mathbf{IXX}\}$ corrects single phase errors $\{\mathbf{E}_a\} = \{\mathbf{ZII}, \mathbf{IZI}, \mathbf{IIZ}\}$.

$$\langle \mathbb{S} \rangle = \{\mathbf{ZZI|III|III}, \mathbf{IZZ|III|III}, \mathbf{III|ZZI|III}, \mathbf{III|IZZ|III}, \mathbf{III|III|ZZI}, \mathbf{III|III|IZZ}, \mathbf{XXX|XXX|III}, \mathbf{III|XXX|XXX}\}$$

Shor code corrects single qubit errors $\{\mathbf{E}_a\} = \{\text{any single qubit error}\}$

$\langle \mathbb{S} \rangle = \{\mathbf{XZZXI}, \mathbf{IXZZX}, \mathbf{XIXZZ}, \mathbf{ZXIXZ}\}$

Five bit code corrects single qubit errors $\{\mathbf{E}_a\} = \{\text{any single qubit error}\}$

All encode one qubit.

$\langle \mathbb{S} \rangle = \{\mathbf{ZZZZZZZZ}, \mathbf{XXXXXXXX}, \mathbf{IXIXYZYZ}, \mathbf{IXZYIXZY}, \mathbf{IYXZIYXZ}\}$

8 qubit code corrects single qubit errors $\{\mathbf{E}_a\} = \{\text{any single qubit error}\}$

. . . but encodes 3 qubits!

# Pauli Want a Qubit?

Return to our old friend the bit flip code: $\mathbf{S}_1 = \mathbf{ZZI}, \mathbf{S}_2 = \mathbf{IZZ}$

$$C = \text{Span}[|000\rangle, |111\rangle]$$

How do we perform encoded opeator on code?

Example: encoded "$\mathbf{X}$": $|000\rangle \Leftrightarrow |111\rangle$

$$\begin{array}{lllllllll} \mathbf{XXX}|000\rangle & = & |111\rangle & -\mathbf{XYY}|000\rangle & = & |111\rangle & -\mathbf{YXY}|000\rangle & = & |111\rangle \\ \mathbf{XXX}|111\rangle & = & |000\rangle & -\mathbf{XYY}|111\rangle & = & |000\rangle & -\mathbf{YXY}|111\rangle & = & |000\rangle \end{array}$$

What is interesting about these operators $(\mathbf{XXX}, \mathbf{XYY}, \mathbf{YXY})$?

They commute with all stabilizer group elements . . .

. . . but are not themselves elements of the stabilizer group.

$$\mathrm{S} = \{\mathbf{III}, \mathbf{ZZI}, \mathbf{IZZ}, \mathbf{ZIZ}\}$$

$(\mathbf{XXX})\mathbf{S}$, $\mathbf{S} \in \mathrm{S}$ acts on code same as $\mathbf{XXX}$

Modulo the stabilizer $\mathbf{S}$

$\mathbf{S}_1 = \mathbf{ZZI}, \mathbf{S}_2 = \mathbf{IZZ}$

Logical opeators: $\bar{\mathbf{X}} = \mathbf{XXX}$, $\bar{\mathbf{Y}} = \mathbf{YXX}$, $\bar{\mathbf{Z}} = \mathbf{ZII}$

$$\mathbf{XXX}|000\rangle = |111\rangle \qquad \mathbf{ZII}|000\rangle = |000\rangle$$
$$\mathbf{XXX}|111\rangle = |000\rangle \qquad \mathbf{ZII}|111\rangle = -|111\rangle$$

Modulo the stabilizer **S**

Feel free to randomly insert "Modulo the stabilizer **S**" into my sentence

Logical operators used to define "encoded" computational basis $\pm 1$ eigenvalues of $\bar{Z}$.

## Logical Operators for a Stabilizer Code

Given a stabilizer group $\mathbb{S}$, the set of Pauli operators which commute with all of the elements of $\mathbb{S}$, $\{\mathbf{L}|[\mathbf{L}, \mathbf{S}], \forall \mathbf{S} \in \mathbb{S}\}$, for the logical operators (MTS) for the stabilizer code.

$\mathbf{S}(\mathbf{L}|\psi_C\rangle) = \mathbf{L}(\mathbf{S}|\psi_C\rangle) = (\mathbf{L}|\psi_C\rangle$: $\mathbf{L}$ preserves codespace.

# So Far

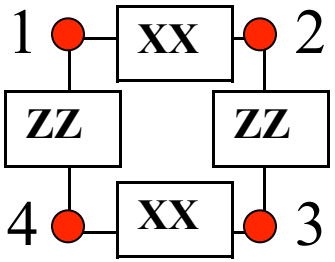
- Introduction to Quantum Error Correction
- Stabilizer Codes

# Next

- Quantum Error Correction in Physical Systems

# Supercoherence

4 qubit system




Stabilizer :

$$S_1 = XXXX$$

$$S_2 = ZZZZ$$

Single qubit error detecting code

Encoded operations:

$$X_1 = XXII,\ Z_1 = IZZI$$

$$X_2 = IXXI,\ Z_2 = ZZII$$

System Hamiltonian:

$$H = XXII + IIXX + ZIIZ + IZZI$$

Note symmetries: $[H, S_i] = [H, X_2] = [H, Z_2] = 0$

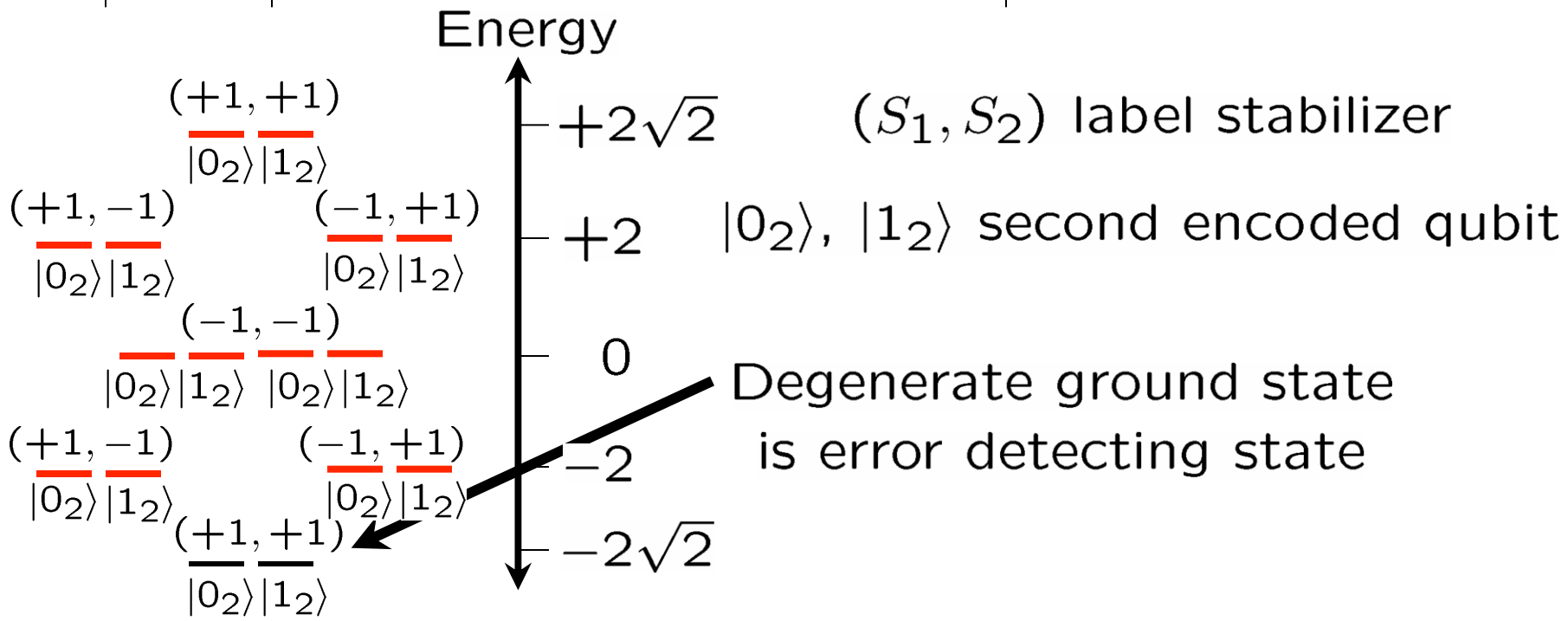
$$H = X_1(I + S_1) + Z_1(I + S_2)$$

Does not involve 2nd encoded qubit.

$\Downarrow$

every energy level at least two-fold degenerate

| $S_1$ | $S_2$ | $\mathbf{H} = \mathbf{X}_1(1+S_1) + \mathbf{Z}_1(1+S_2)$ | Eigenvalues |
|---|---|---|---|
| $+1$ | $+1$ | $2(\mathbf{X}_1 + \mathbf{Z}_1)$ | $\pm 2\sqrt{2}$ |
| $+1$ | $-1$ | $2\mathbf{X}_1$ | $\pm 2$ |
| $-1$ | $+1$ | $2\mathbf{Z}_1$ | $\pm 2$ |
| $-1$ | $-1$ | $\mathbf{0}$ | 0 |



# Supercoherence

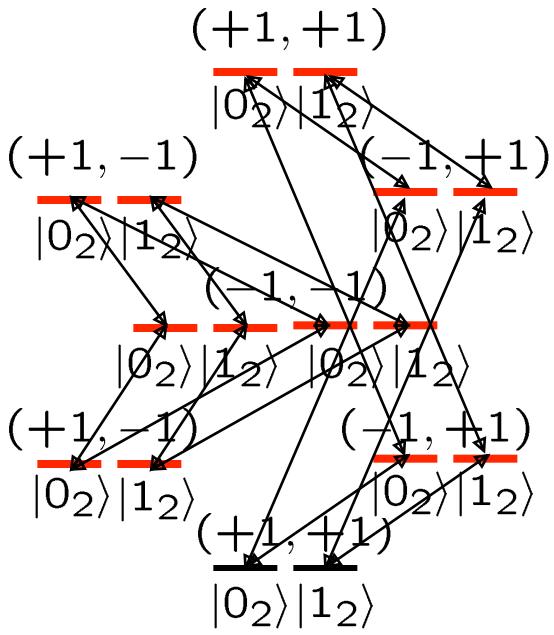
$$\mathbf{H} = \mathbf{X}_1(1 + S_1) + \mathbf{Z}_1(1 + S_2)$$

$\mathbf{S}_i\mathbf{P} = -\mathbf{P}\mathbf{S}_i$ for some $i$ $\quad$ $\mathbf{P}$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

Effect of **ZIII**

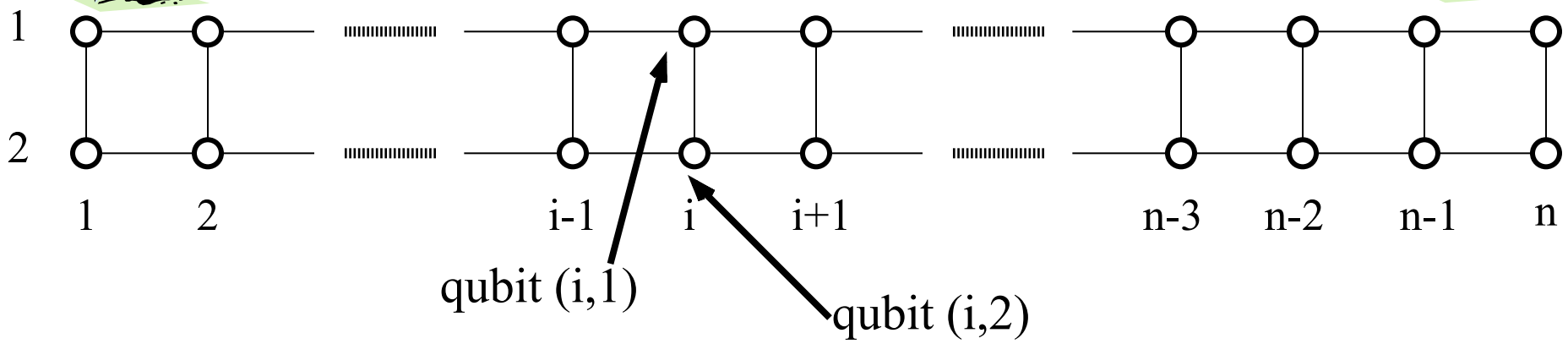



$$\underline{(+1,}\ \underline{+1)}$$
$$|0_2\rangle \qquad\quad |1_2\rangle$$

✗ **FORBIDDEN**

D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

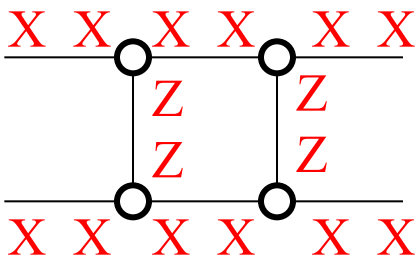
# Ladder Qubits




$\sigma_\alpha^{(i,j)} \equiv \alpha^{\text{th}}$ Pauli operator acting on the $(i, j)$ qubit tensored with $I$ on all other qubits

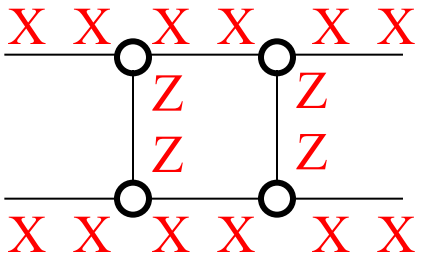
The Hamiltonian

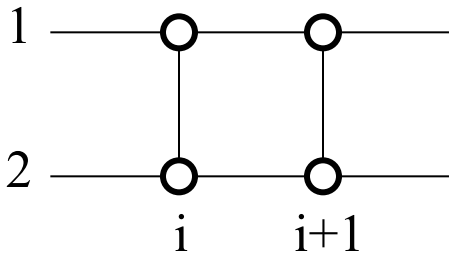
$$H = -\omega \left[ \sum_{i=1}^{n} \sigma_Z^{(i,1)} \sigma_Z^{(i,2)} + \sum_{i=1}^{n-1} \left( \sigma_X^{(i,1)} \sigma_X^{(i+1,1)} + \sigma_X^{(i,2)} \sigma_X^{(i+1,2)} \right) \right]$$




D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

# Basis Change




New basis:

$$\mathbf{X}_1^{(i)} = \sigma_x^{(i,1)}, \mathbf{Z}_1^{(i)} = \sigma_z^{(i,1)}\sigma_z^{(i,2)}$$

$$\mathbf{X}_2^{(i)} = \sigma_x^{(i,1)}\sigma_x^{(i,2)}, \mathbf{Z}_2^{(i)} = \sigma_z^{(i,2)}$$

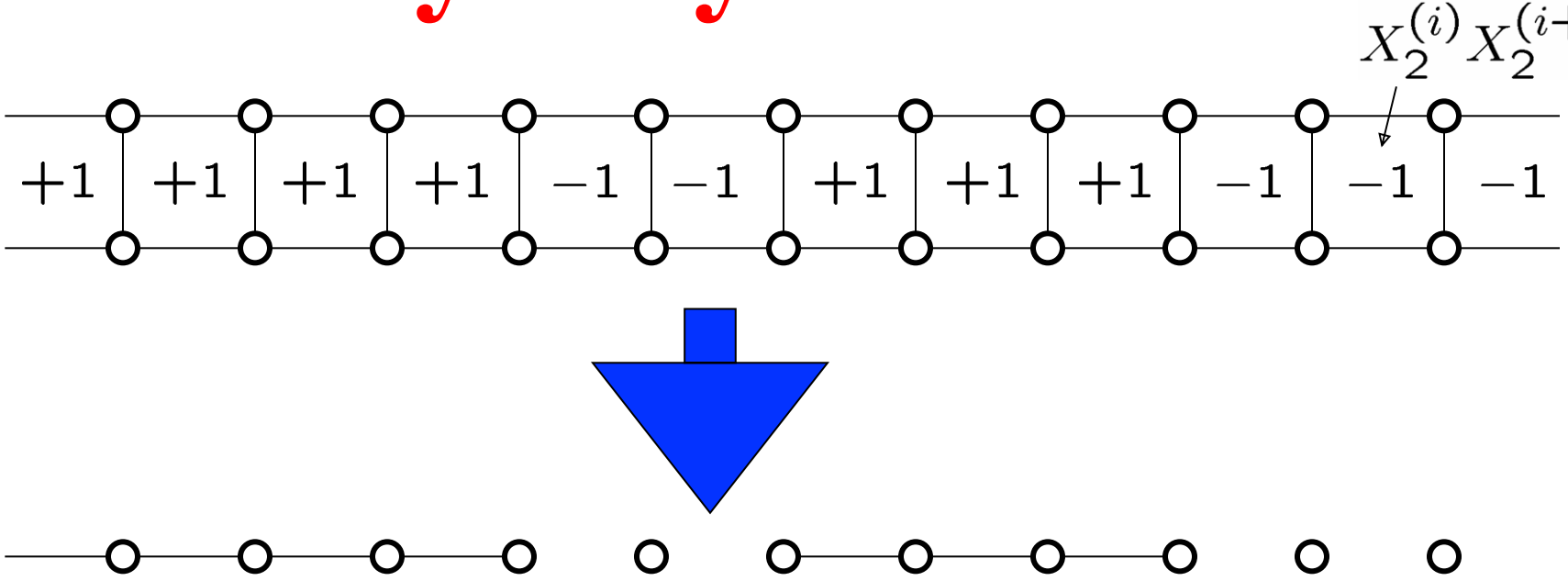
$$\mathbf{H} = -\omega\left(\sum_{i=1}^{n}\mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1}\mathbf{X}_1^{(i)}\mathbf{X}_1^{(i+1)}\left(\mathbf{I} + \mathbf{X}_2^{(i)}\mathbf{X}_2^{(i+1)}\right)\right)$$

basis $|\vec{Z}_1, \vec{X}_2\rangle = |Z_1^{(1)}, \ldots, Z_1^{(n)}, X_2^{(1)}, \ldots, X_2^{(n)}\rangle$ $\quad Z_1^{(i)}, X_2^{(i)} \in \{\pm 1\}$

$$\mathbf{H} = -\omega\left(\bigoplus_{\vec{X}_2}\left(\sum_{i=1}^{n}\mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1}c_i(\vec{X}_2)\mathbf{X}_1^{(i)}\mathbf{X}_1^{(i+1)}\right)\otimes|\vec{X}_2\rangle\langle\vec{X}_2|\right)$$

$$c_i(\vec{X}_2) = (1 + X_2^{(i)}X_2^{(i+1)})$$

# Everybody's Favorite




Ising + transverse field

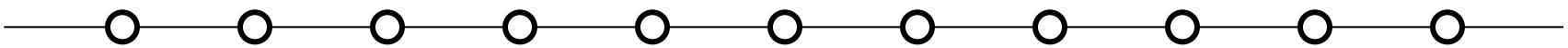
transverse field only

Ising + transverse field

transverse field only

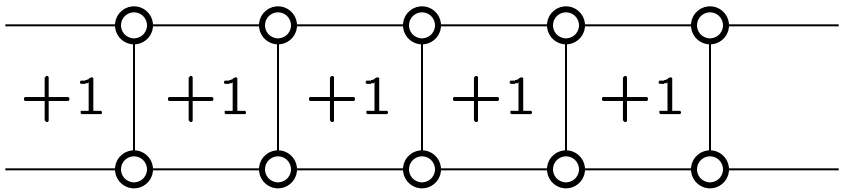
$$\mathbf{H} = -\omega \left( \sum_{i=1}^{n} \mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1} c_i(\vec{X}_2) \mathbf{X}_1^{(i)} \mathbf{X}_1^{(i+1)} \right) \quad c_i(\vec{X}_2) \in \{0, 2\}$$

# Grounded



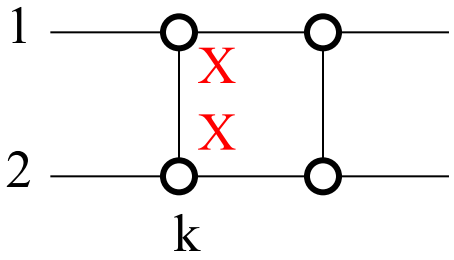
$$c_i(\vec{X}_2) = 2, \forall i$$

Ground state is two-fold degenerate




$$\mathbf{X}_2^{(i)} = +1, \forall k$$

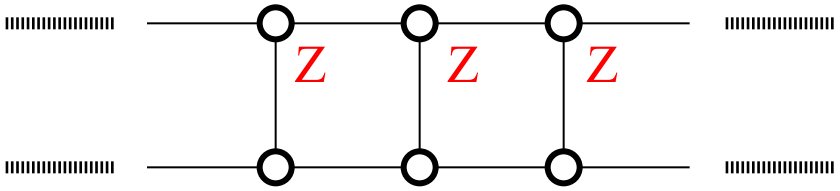
$$\mathbf{X}_2^{(i)} = -1, \forall k$$




$$\bar{\mathbf{X}} = \mathbf{X}_2^{(k)}$$

Encoded operations

$$[\mathbf{H}, \bar{\mathbf{X}}] = [\mathbf{H}, \bar{\mathbf{Z}}] = 0$$




$$\bar{\mathbf{Z}} = \prod_{i=1}^{n} \mathbf{Z}_2^{(i)}$$

# Detect and Correct

$$(\mathbb{C}^2)^{2n} = \mathcal{H} = \mathcal{H}_s \otimes \mathcal{H}_h \otimes \mathcal{H}_l$$

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

$\mathcal{H}_l = (\mathbb{C}^2)$ logical qubit

$\mathcal{H}_h = (\mathbb{C}^2)^n$ Hamiltonian space

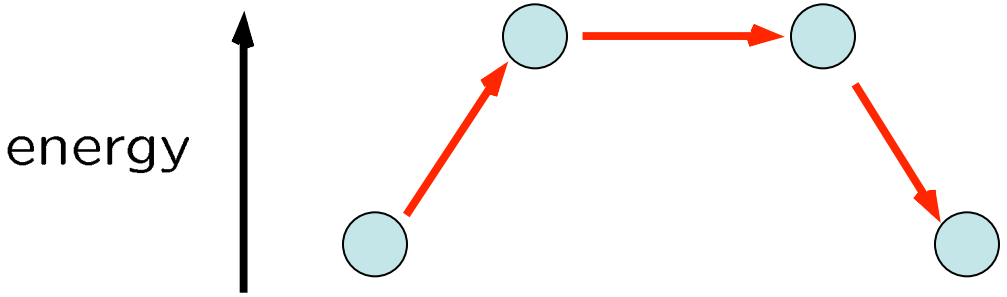
$\mathcal{H}_s = (\mathbb{C}^2)^{n-1}$ stabilizer code

$\bar{\mathbf{X}}, \bar{\mathbf{Z}} \in \mathcal{B}_l \quad \mathbf{X}_2^{(i)}\mathbf{X}_2^{(i+1)} \in \mathcal{B}_s \quad \mathbf{Z}_1^{(i)}, \mathbf{X}_1^{(i)} \in \mathcal{B}_h$

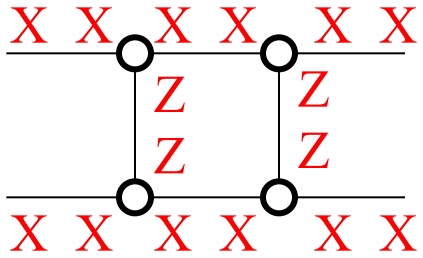
Elements of $\mathcal{B}_s$ imply distance $n$ qubit $\sigma_z$ detecting code

Can correct $\left\lfloor \frac{n-1}{2} \right\rfloor$, $\sigma_z$ errors

but only 1 qubit $\sigma_x$ detecting code

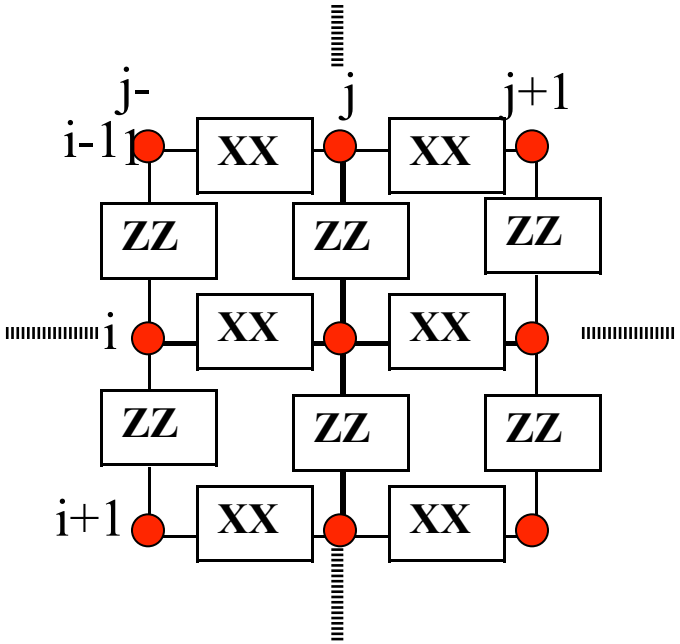



disorders at finite temperature



# Into the 2nd

$$\mathbf{H} = \sum_{i=1}^{n} \sum_{j=1}^{n-1} \sigma_x^{(i,j)} \sigma_x^{(i,j+1)} + \sum_{i=1}^{n-1} \sum_{j=1}^{n} \sigma_z^{(i,j)} \sigma_z^{(i+1,j)}$$




$$[\mathbf{H}, \bar{\mathbf{X}}] = [\mathbf{H}, \bar{\mathbf{Z}}] = 0$$

Look at group algebra generated by individual bounds

Abelian invariant algebra generated double rows:

$$\mathbf{Z}_k = \prod_{j=1}^{n} \sigma_z{}^{(k,j)} \sigma_z^{(k+1,j)} \qquad \mathbf{X}_k = \prod_{i=1}^{n} \sigma_x{}^{(i,k)} \sigma_x^{(i,k+1)}$$

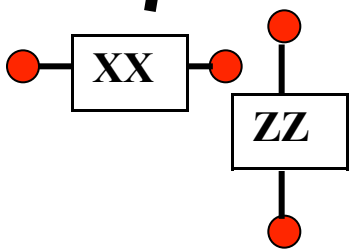
Nonabelian commutant: $\bar{\mathbf{Z}} = \prod_{j=1}^{n} \sigma_z{}^{(k,j)}$ $\bar{\mathbf{X}} = \prod_{i=1}^{n} \sigma_x{}^{(i,k)}$

# Tic-Tac-"DOH!"

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

$$\mathbf{Z}_k = \prod_{j=1}^{n} {\sigma_z}^{(k,j)} {\sigma_z}^{(k+1,j)}$$

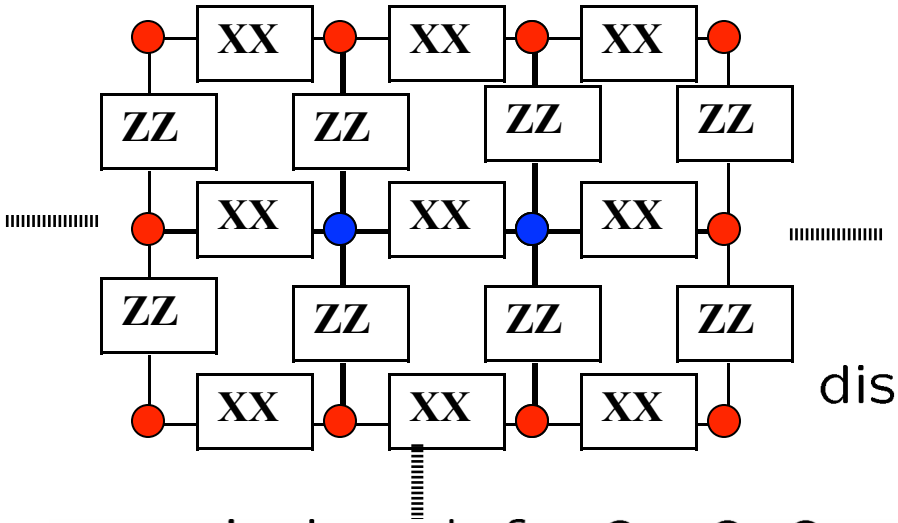
$$\mathbf{X}_k = \prod_{i=1}^{n} {\sigma_x}^{(i,k)} {\sigma_x}^{(i,k+1)}$$

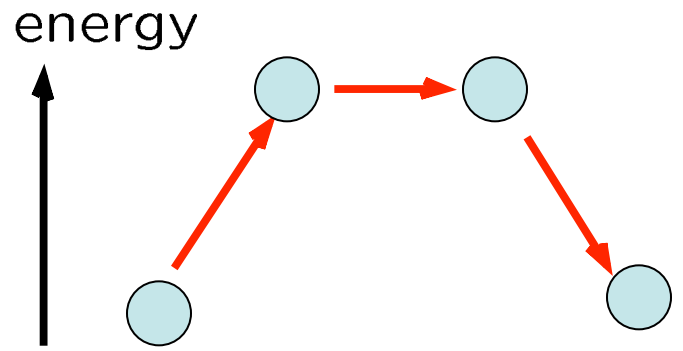



$$\bar{\mathbf{Z}} = \prod_{j=1}^{n} {\sigma_z}^{(l,j)}$$

$$\bar{\mathbf{X}} = \prod_{i=1}^{n} {\sigma_x}^{(i,l)}$$

$\sigma_z$ errors

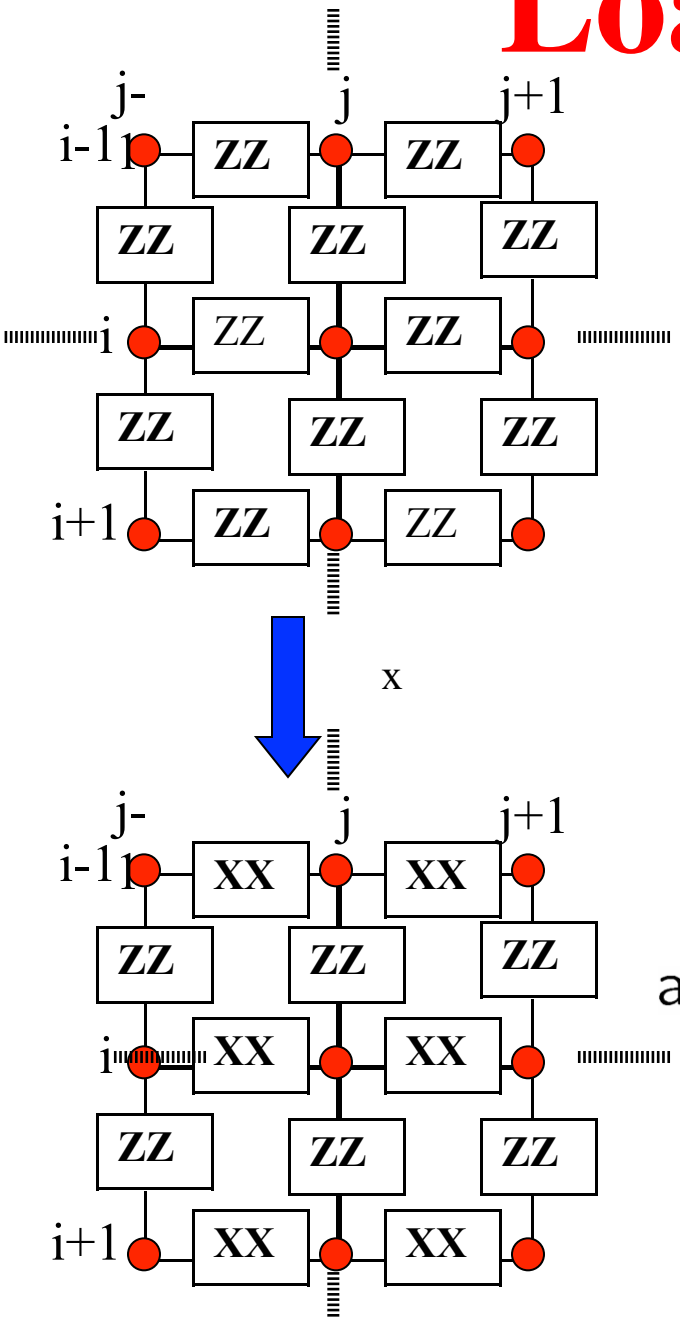


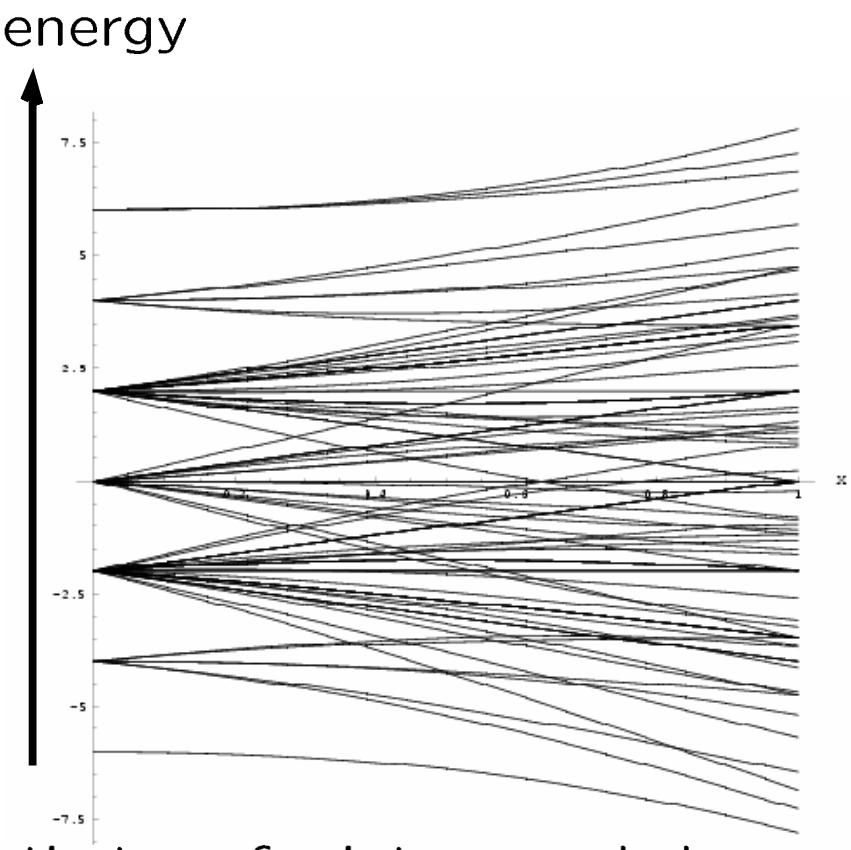




disorders at finite temperature

numerical work for $3 \times 3$, $3 \times 4$, $4 \times 4$, $4 \times 5$ supports

# Loading 3x3



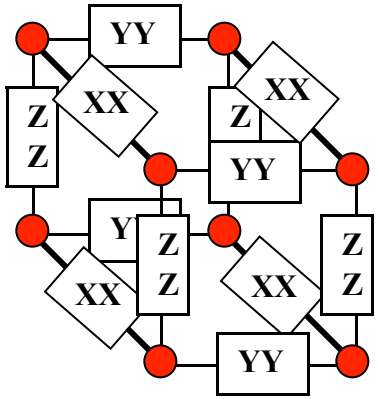




adiabatic transfer into encoded ground state

# Into the Third Dimension

$n \times n \times n$

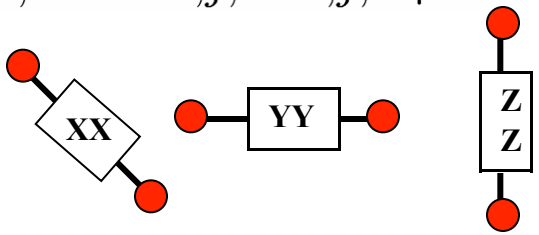
$n$ odd



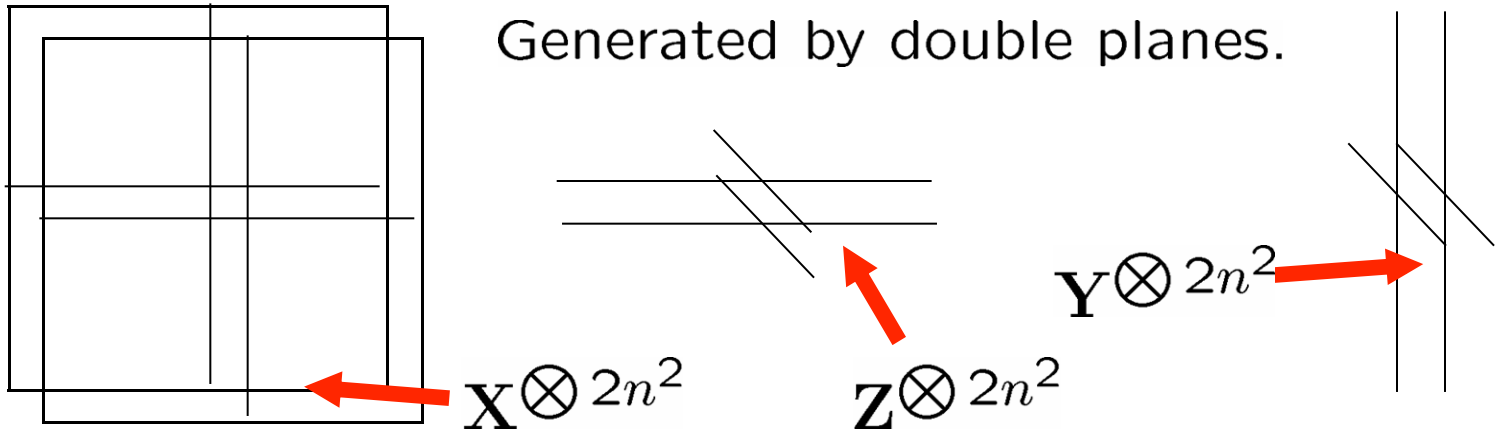

Cubic lattice with qubits at vertices

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

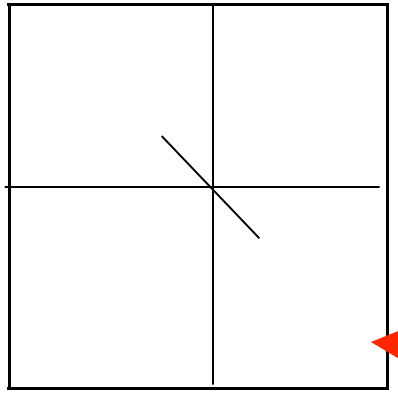
Look at group algebra generated by bonds:




Abelian invariant subalgebra:



# Logical Operators



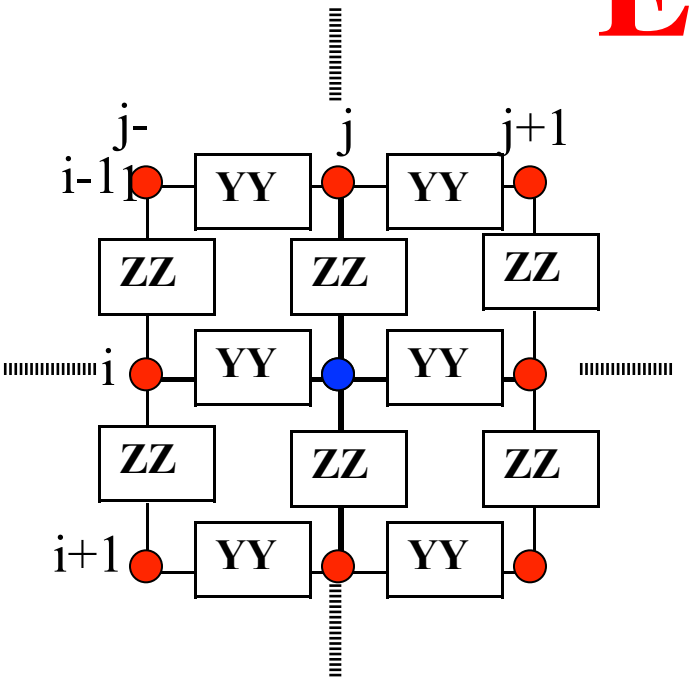
$$\bar{\mathbf{Z}} = \mathbf{Z}^{\otimes n^2}$$

$$\bar{\mathbf{X}} = \mathbf{X}^{\otimes n^2}$$

$$\bar{\mathbf{Y}} = \mathbf{Y}^{\otimes n^2}$$

$$[\bar{\mathbf{X}}, \mathbf{H}] = [\bar{\mathbf{Y}}, \mathbf{H}] = [\bar{\mathbf{Z}}, \mathbf{H}] = 0$$

# Errors




$\sigma_z$ error

$\sigma_y$ errors stringlike

$\sigma_x$ errors stringlike

$\sigma_z$ errors stringlike

# Order Parameter and Beyond

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

stabilizer — Hamiltonian — Logical qubit

Subsystem code:

error ⇓ diagnose and fix

Error in $\mathcal{B}$ ⇓ makes effect idenity on $\mathcal{B}_s, \mathcal{B}_l$

error left on $\mathcal{B}_h$
don't care (tensor product)

This allows us to write down a subsystem order parameter.

Current project: Quantum Monte Carlo for this order parameter.



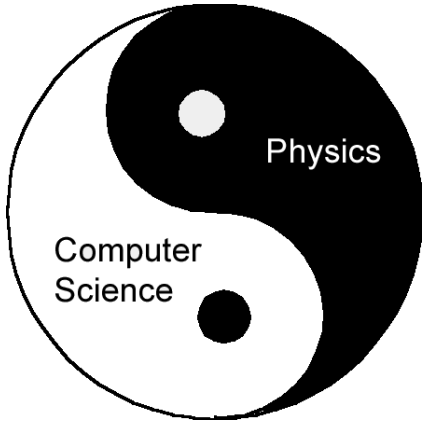
**Rolf Landauer**
**IBM**

# Quantum Error Correction!



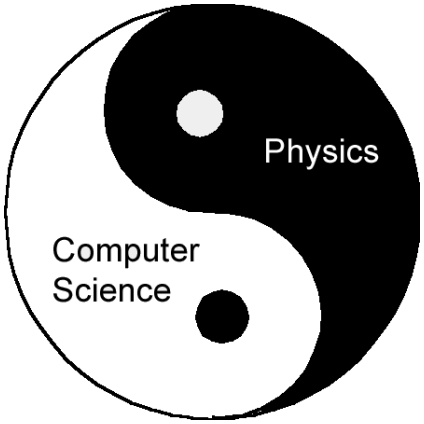
Natural Fault Tolerance

# Fin










# Quantum Computers Which Fix Themselves

**Dave Bacon, Caltech**









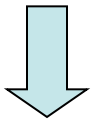
# Unsurfing The Subspaces

Measuring the generators $\mathbf{G}_i$ can be simultaneously performed (they commute!).

Measuring the generators $\mathbf{G}_i$ determines which $s_1, s_2, \ldots, s_k$ subspace the state is in.

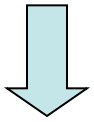
Once the $s_i$ are determined, returning to the codespace is simply rotating back to all $s_i = +1$?

$$|s_1 = +1, s_2 = +1, s_3 = +1, s_4 = +1, m\rangle$$



Error

$$|s_1 = +1, s_2 = -1, s_3 = -1, s_4 = +1, m\rangle$$



Fix ⟵ Correct fix?

$$|s_1 = +1, s_2 = +1, s_3 = +1, s_4 = +1, m\rangle$$

Well yes, from iff condition

# Sizing Up The Code

Let $\mathtt{S}$ generated by $\mathbf{G}_1, \mathbf{G}_2, \ldots, \mathbf{G}_l$. Define the projectors onto the $+1$ eigenspace:

$$\mathbf{P}_i = \frac{1}{2}(\mathbf{I} + \mathbf{G}_i)$$

$$\mathbf{P}_i\mathbf{P}_j = \mathbf{P}_j\mathbf{P}_i, \quad \mathbf{P}_i^2 = \mathbf{P}_i, \quad \mathrm{Tr}[\mathbf{P}_i] = 2^{n-1}$$

Projector onto the codespace is

$$\mathbf{P} = \mathbf{P}_1\mathbf{P}_2\cdots\mathbf{P}_l$$

$$\mathbf{P}^2 = \mathbf{P}, \quad \dim C = \mathrm{Tr}[\mathbf{P}]$$

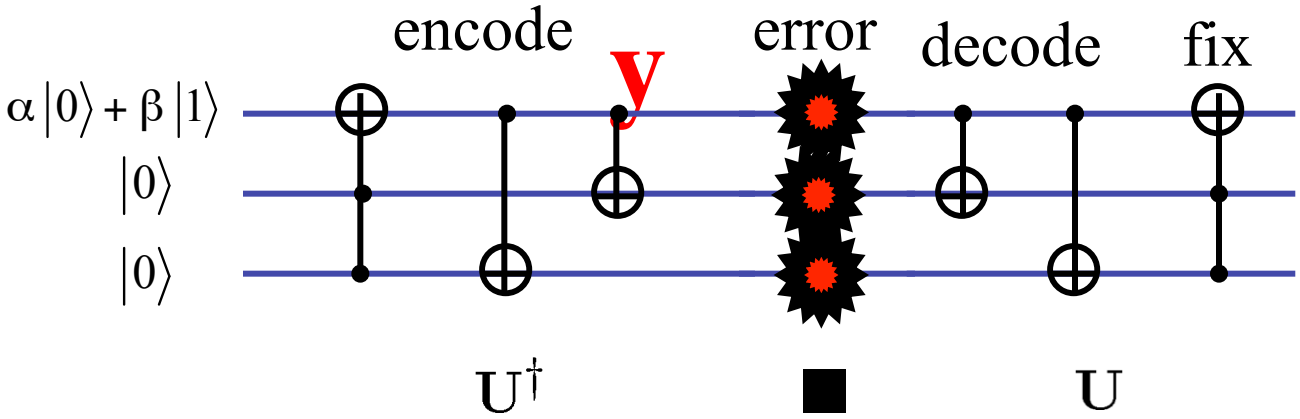
$$\mathrm{Tr}\left[\mathbf{P}_1\mathbf{P}_2\right] = \frac{1}{4}\mathrm{Tr}\left[\mathbf{I} + \mathbf{G}_1 + \mathbf{G}_2 + \mathbf{G}_1\mathbf{G}_2\right] = 2^{n-2}$$

Not identity (generators!)

$$\mathbf{P} = \frac{1}{2^l} \sum_{\alpha_1,\alpha_2,\ldots,\alpha_l \in \{0,1\}} \mathbf{G}_1^{\alpha_1}\mathbf{G}_2^{\alpha_2}\cdots\mathbf{G}_l^{\alpha_l} \Rightarrow \mathrm{Tr}[\mathbf{P}] = 2^{n-l}$$

A stabilizer code generated by $k$ elements has dimension $2^{n-k}$ (encodes $n-k$ qubits)

# Identit




But this is just a basis change: $UEU^\dagger$, $E$ is the error

